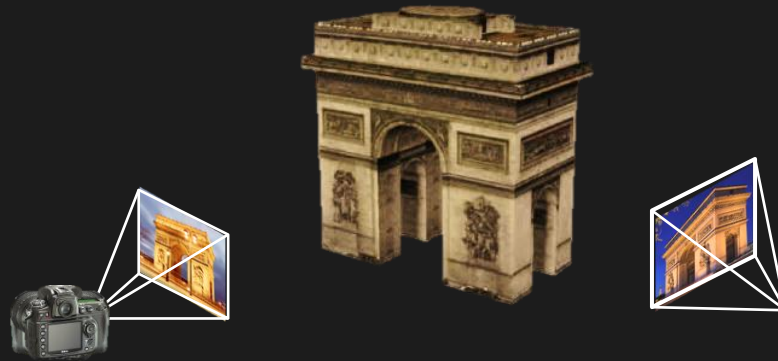


Uncalibrated Binocular Stereo

Computer Vision: AI3619

Uncalibrated Binocular Stereo



Left Image



Right Image

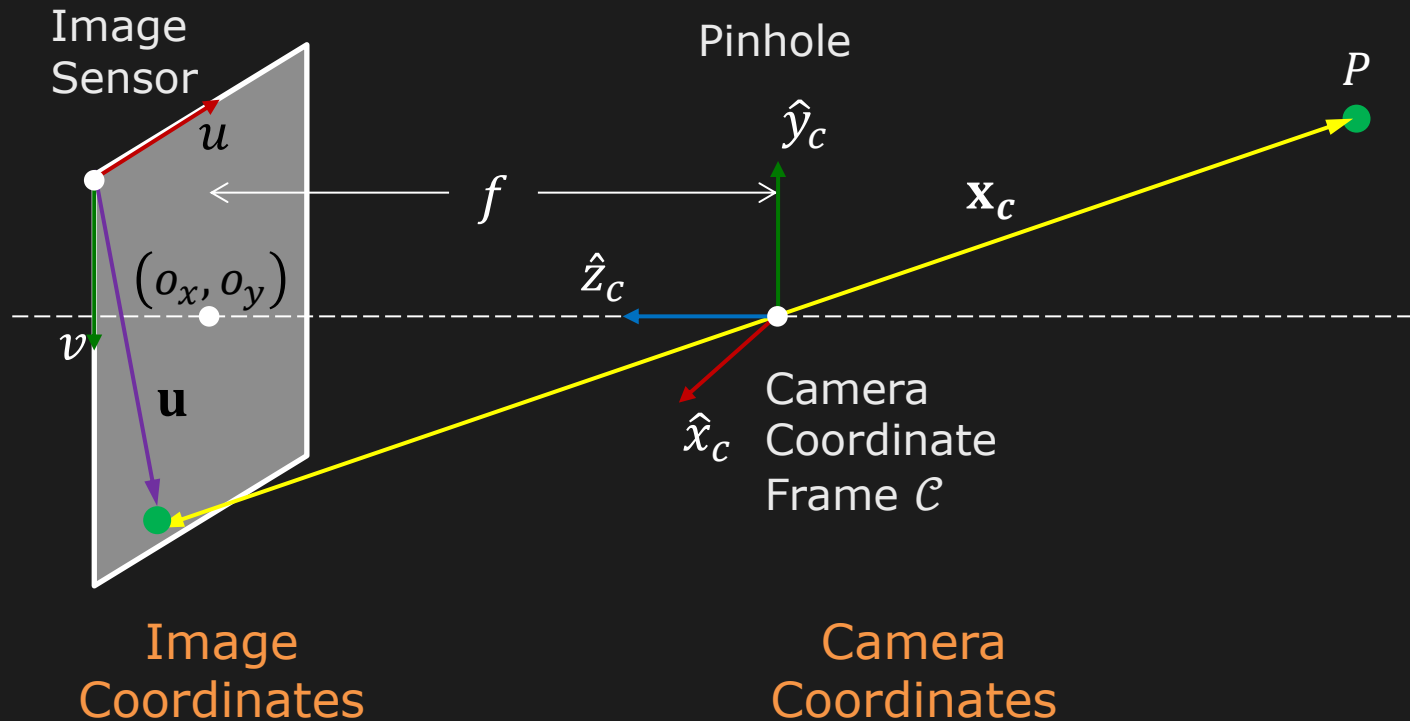
Uncalibrated Binocular Stereo

Method to estimate 3D structure from two arbitrary images of a scene captured with cameras whose intrinsic parameters are known.

Topics:

- (1) Epipolar Geometry
- (2) Essential and Fundamental Matrix
- (3) Stereo Self-Calibration
- (4) Stereopsis

Review: Linear Camera Model



$$\mathbf{u} = \begin{bmatrix} u \\ v \end{bmatrix}$$



$$\mathbf{x}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix}$$

Perspective
Projection

Review: Linear Camera Model

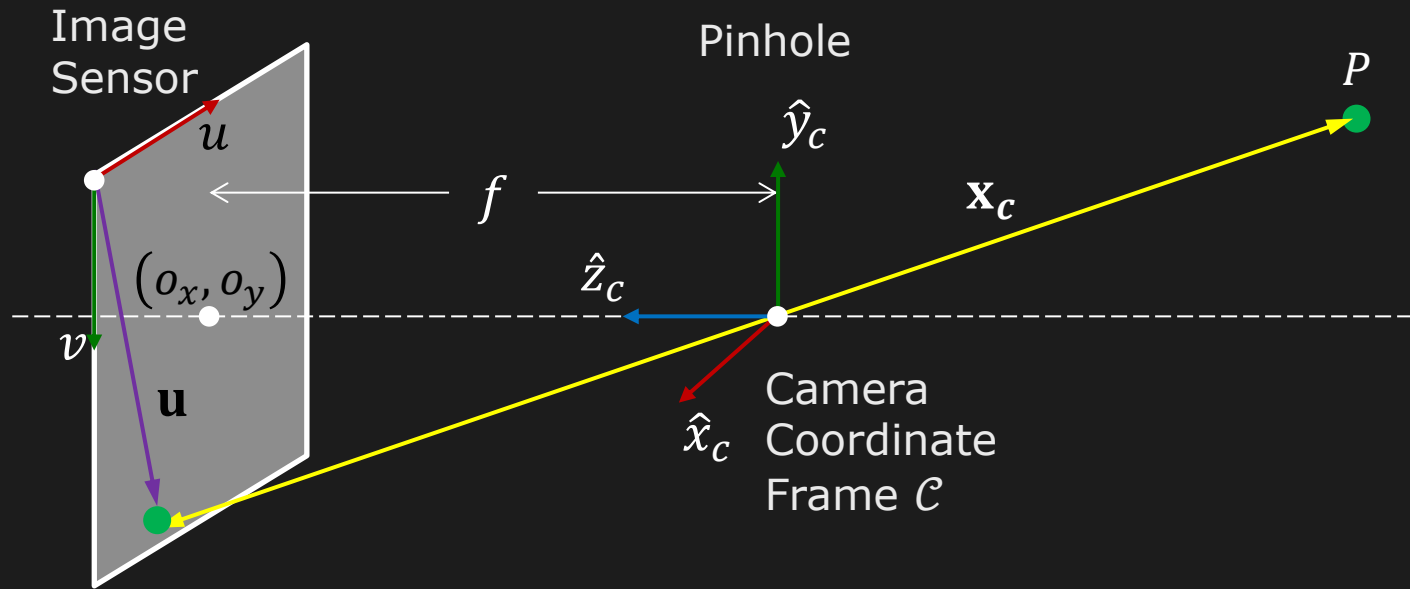


Image
Homogenous
Coordinates

$$\tilde{\mathbf{u}} = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

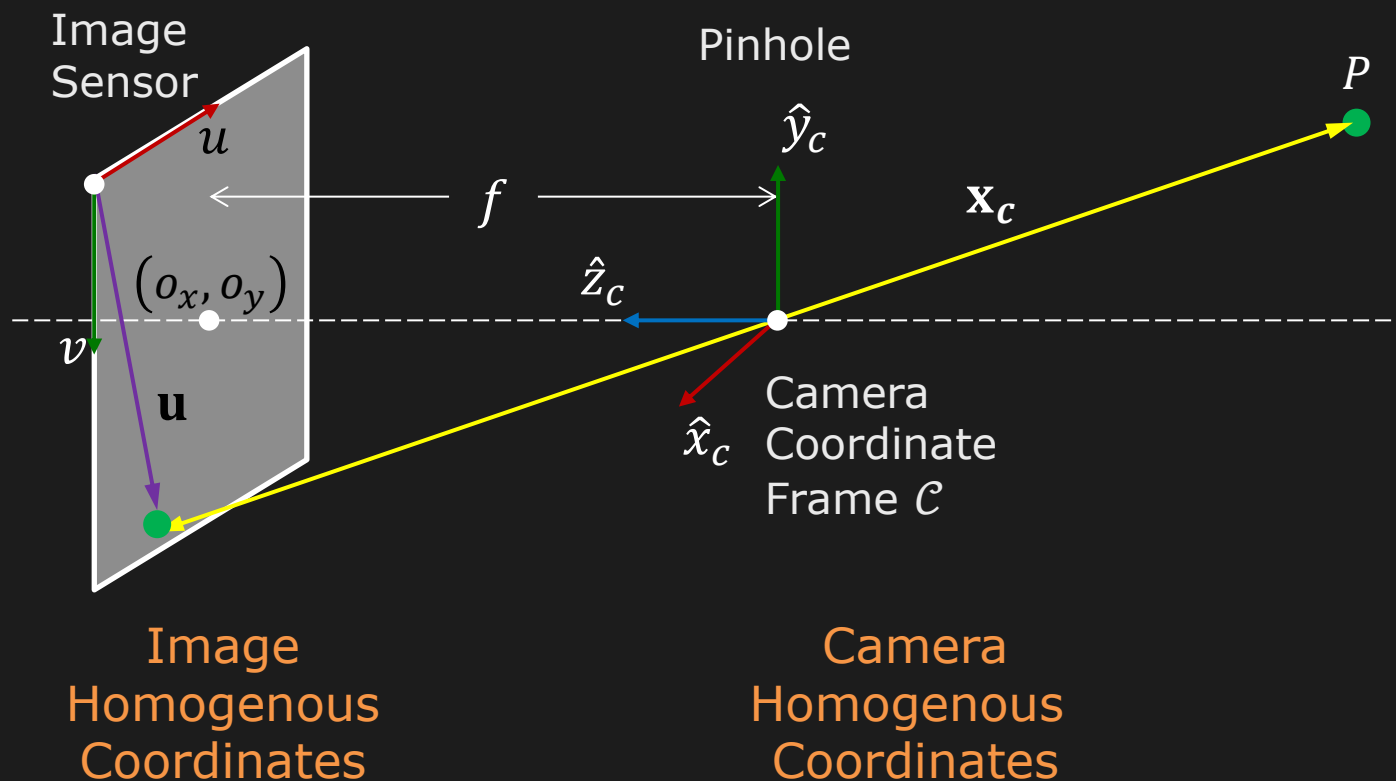
Camera
Homogenous
Coordinates

$$\tilde{\mathbf{x}}_c = \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$



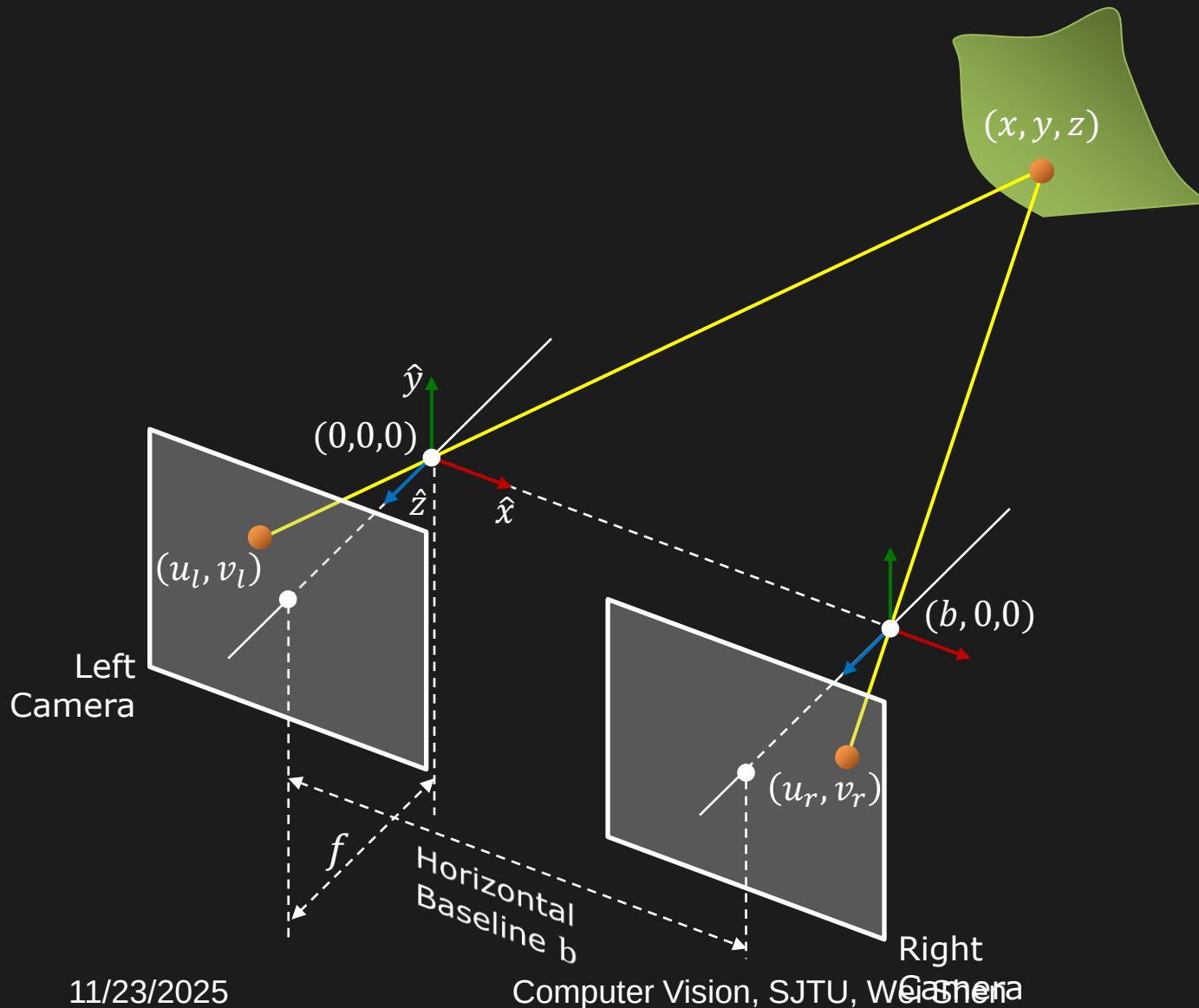
Perspective
Projection

Review: Linear Camera Model

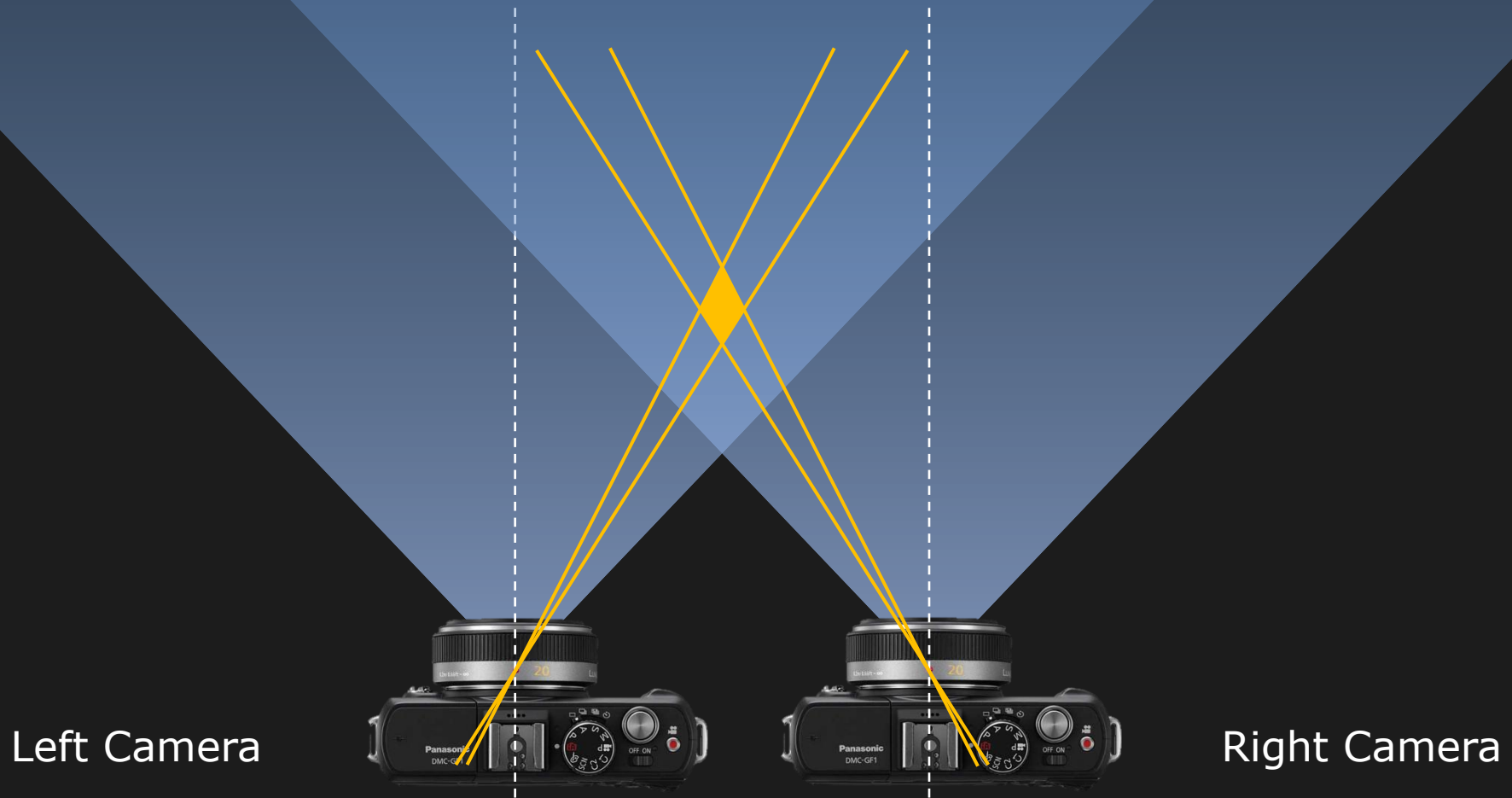


$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} = \begin{bmatrix} f_x & 0 & o_x & 0 \\ 0 & f_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}$$

Review: Simple Stereo

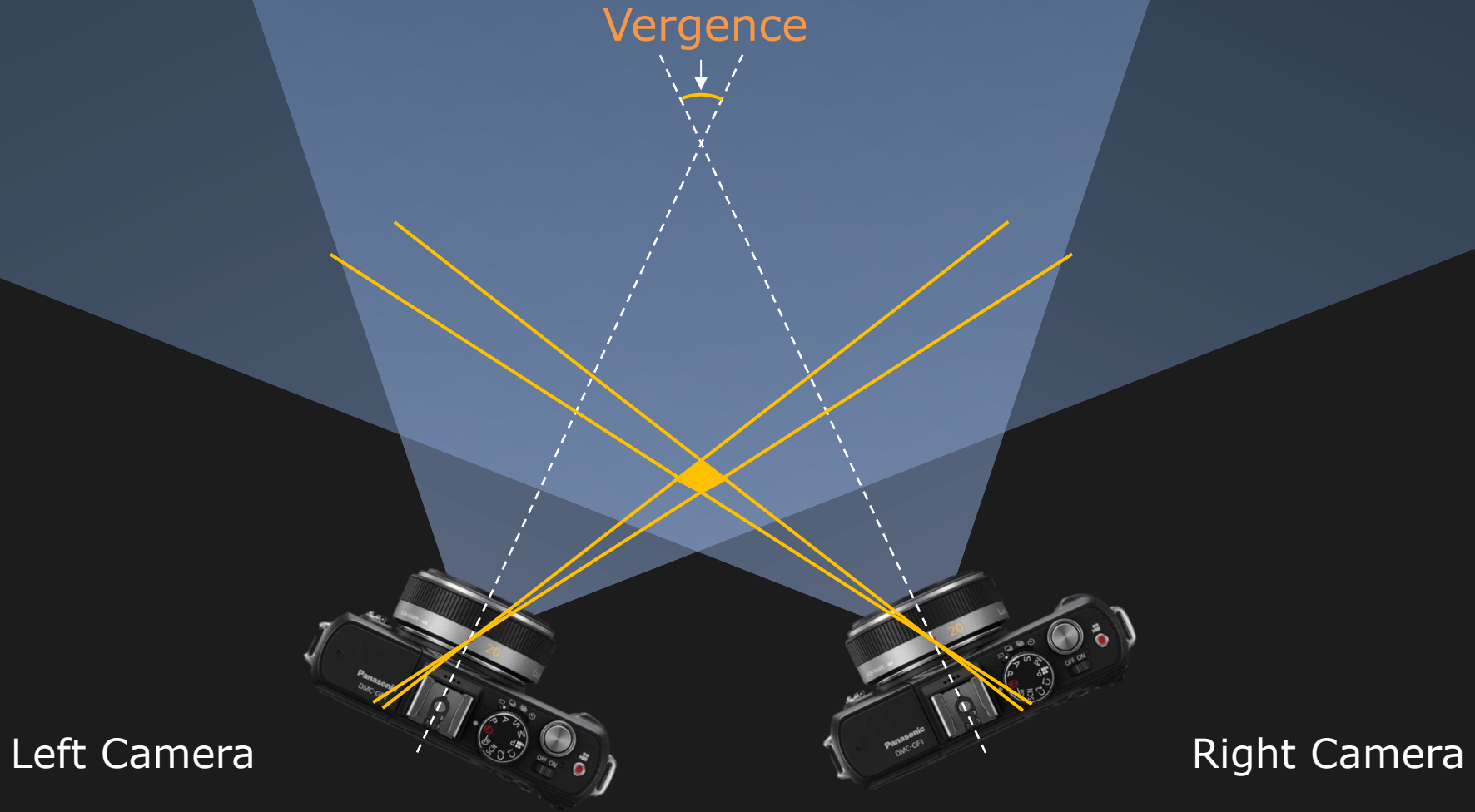


Binocular Field of View



Binocular Field of View is the overlapping field of view.

Binocular Field of View: Vergence



Field of view Decreases; **Accuracy Increases** with Vergence

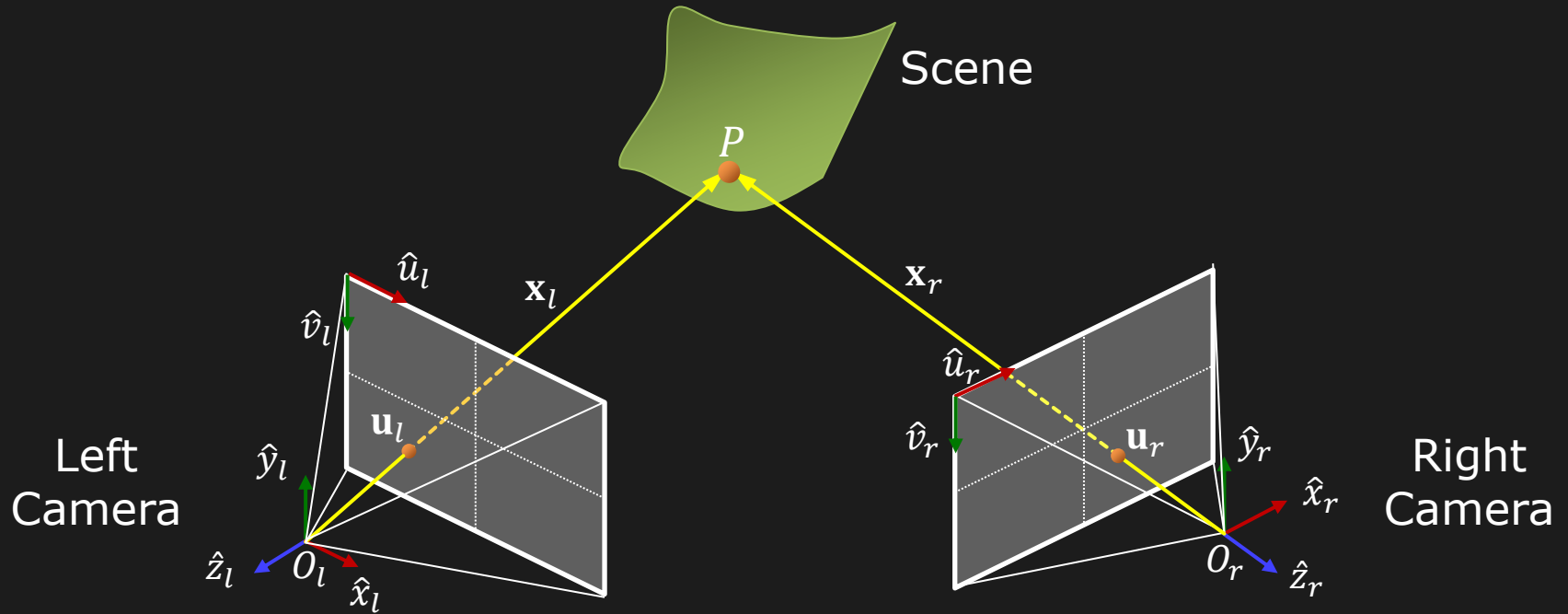
Predator vs. Prey



Predator eyes are optimized for
depth estimation

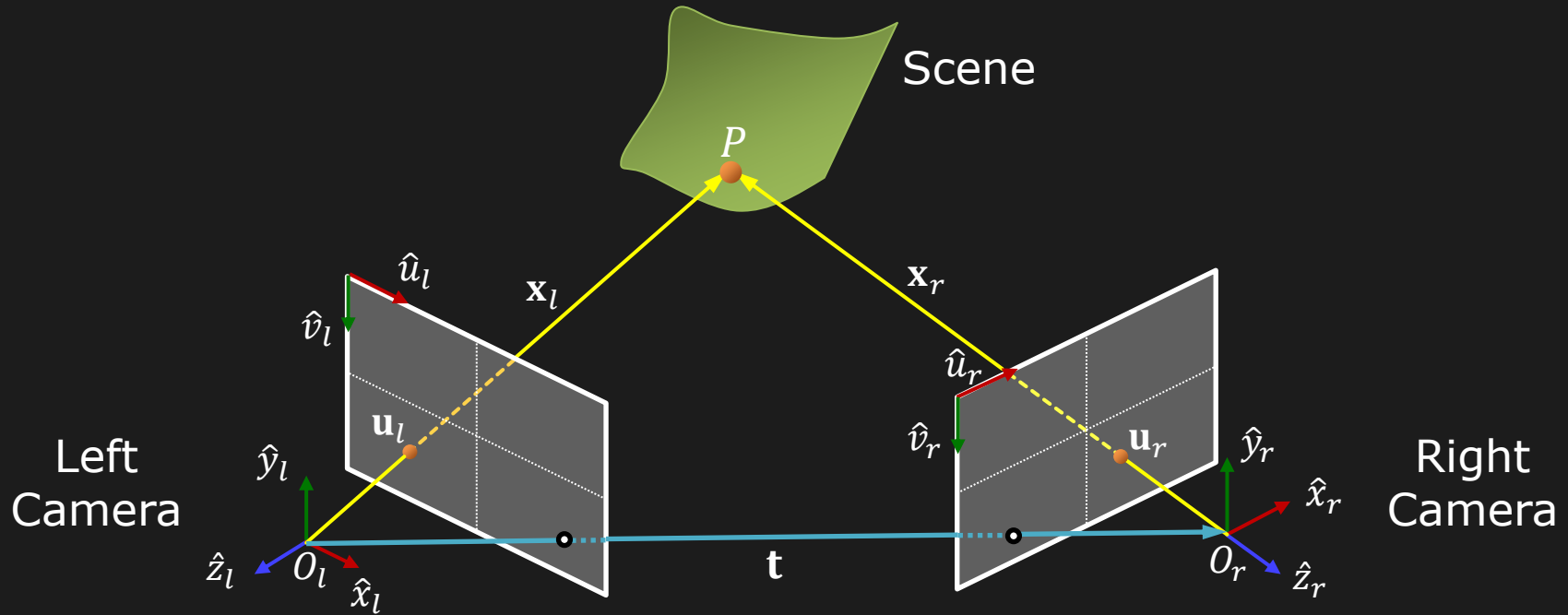
Prey eyes have a large field of view

Uncalibrated Binocular Stereo



Compute depth using two cameras (whose intrinsics are known) with arbitrary position and orientation.

Relative Position and Orientation



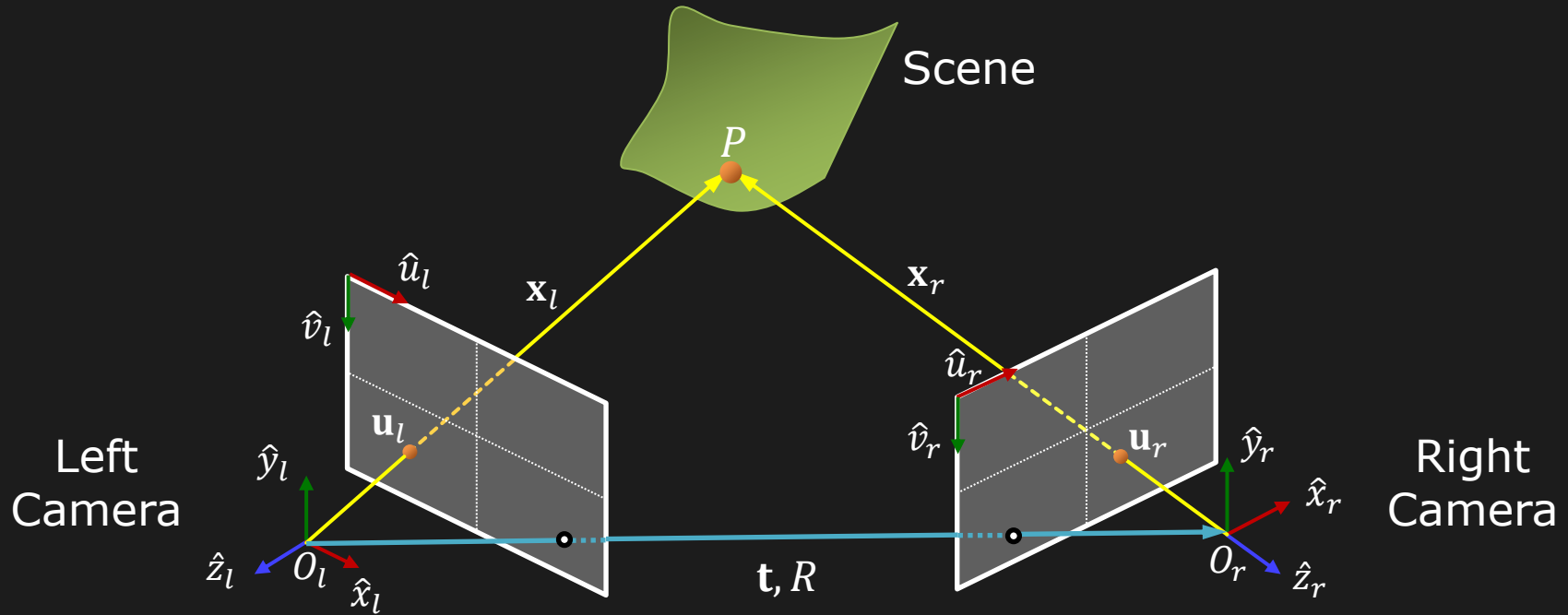
$\mathbf{t}_{3 \times 1}$: Position of Right Camera in Left Camera Coordinate Frame ($\overrightarrow{O_l O_r}$)

$R_{3 \times 3}$: Rotation from Right to Left Camera Coordinate Frame

$$\mathbf{x}_l = R\mathbf{x}_r + \mathbf{t}$$

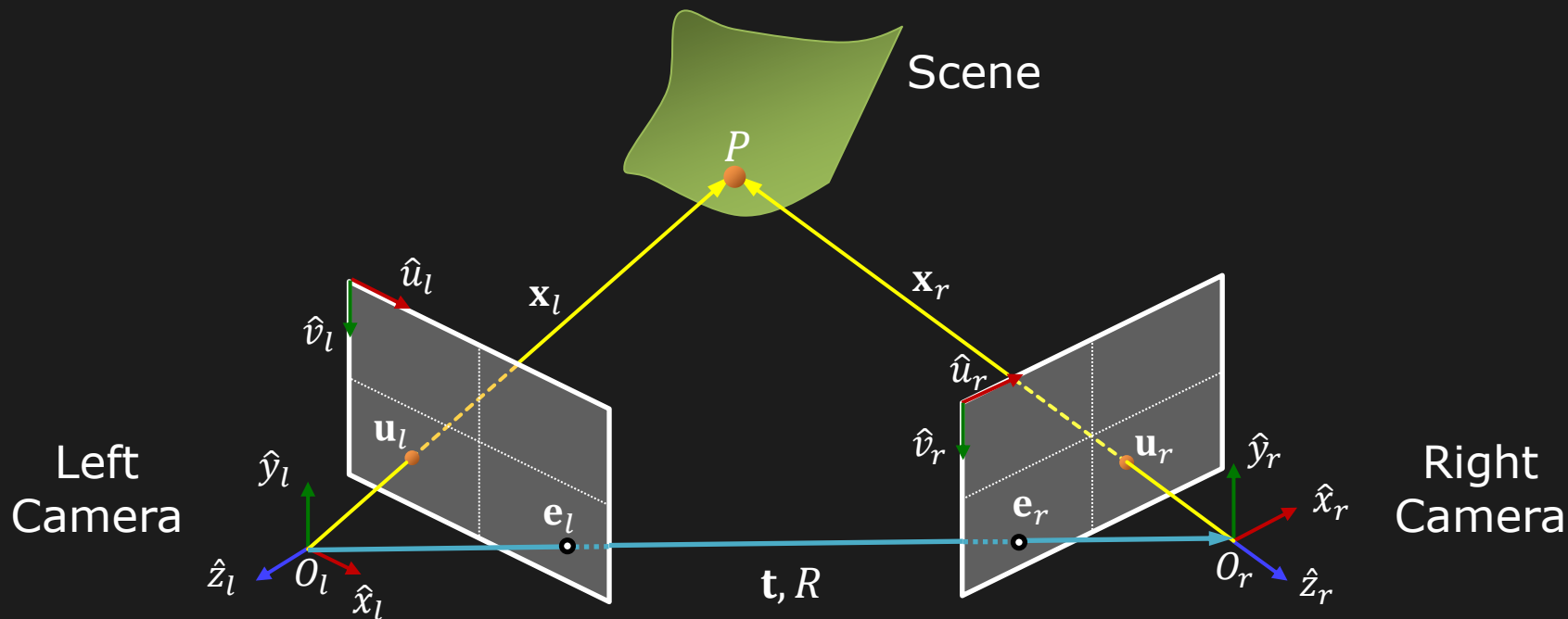
$$\begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Binocular Stereo



1. Assume Camera Intrinsic Parameters f_x, f_y, o_x, o_y are known.
2. Find Relative Camera Position t and Orientation R from the two images.
3. Find Correspondence for each pixel in the two images.
4. Compute Depth for each pixel using Triangulation.

Epipolar Geometry: Epipole

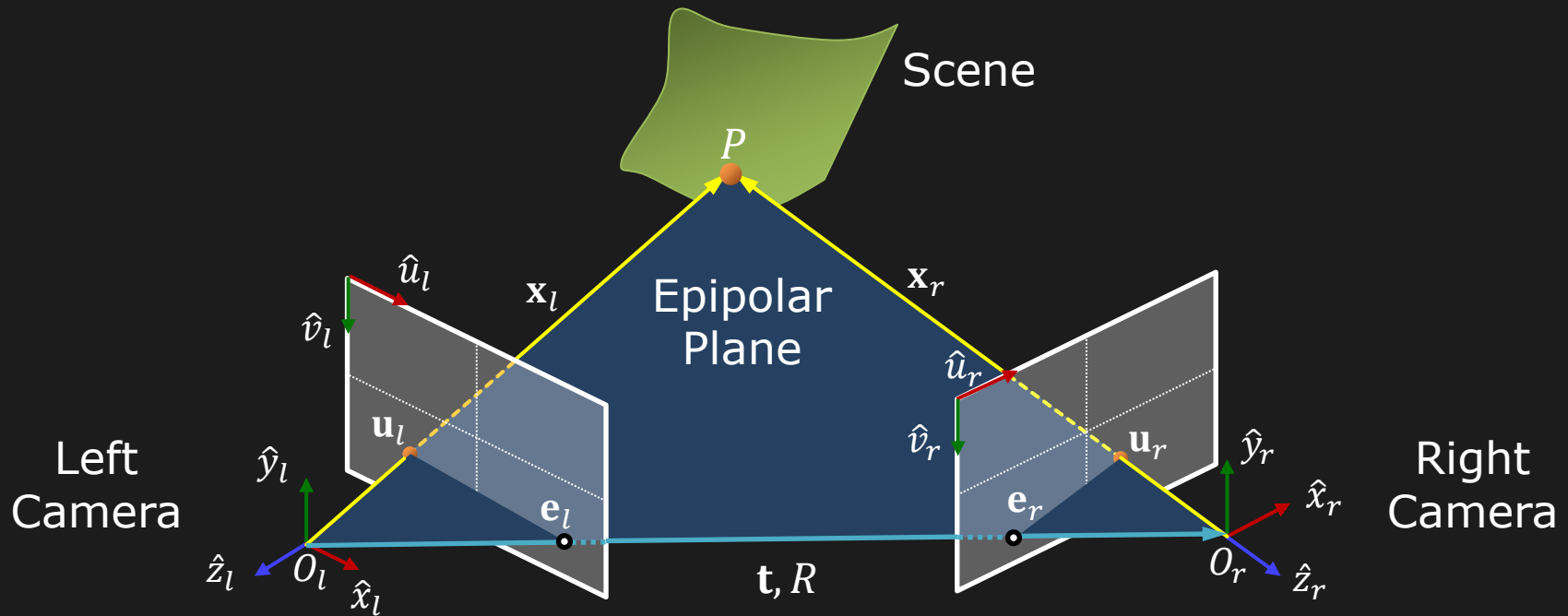


The image point of the origin/pinhole of one camera as viewed by the other camera is called the **epipole**.

e_l and e_r are the epipoles.

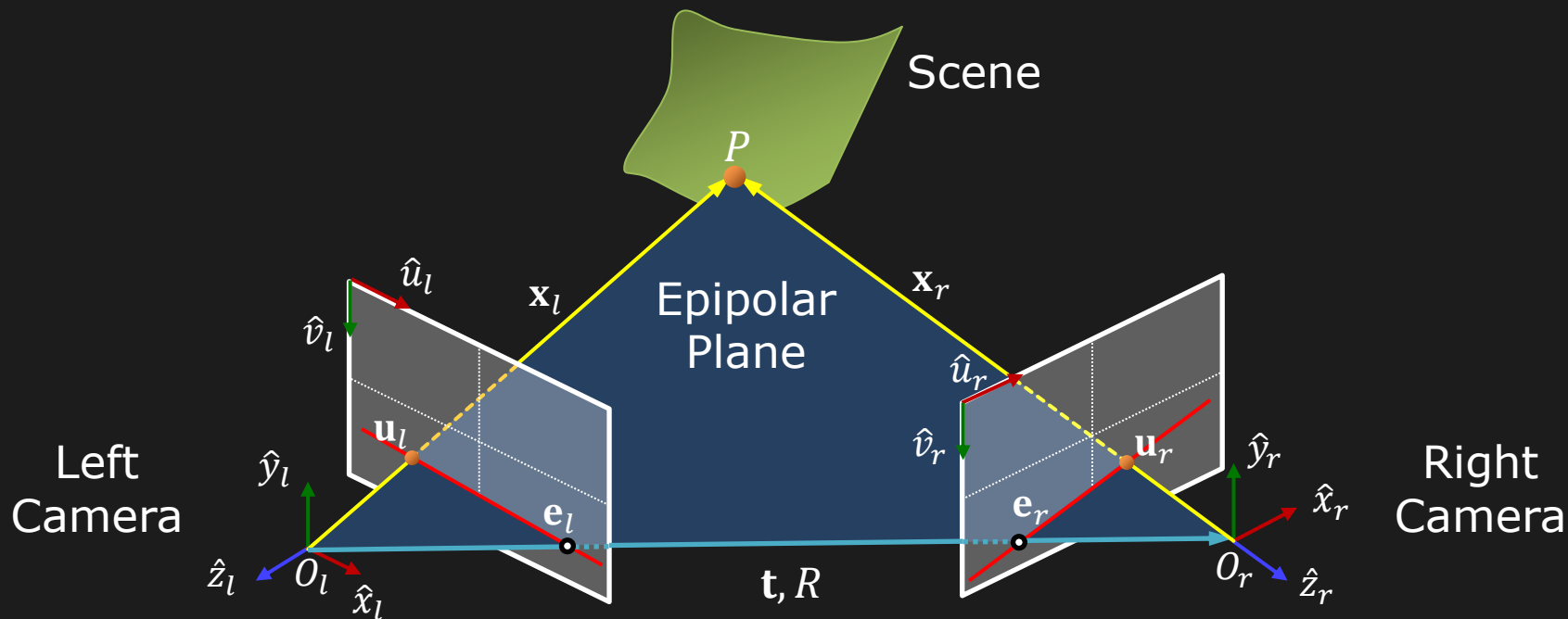
e_l and e_r are unique for a given stereo pair.

Epipolar Geometry: Epipolar Plane



The camera origins (O_l and O_r), the epipoles (e_l and e_r) and any given scene point all lie on a plane called the **Epipolar Plane**.

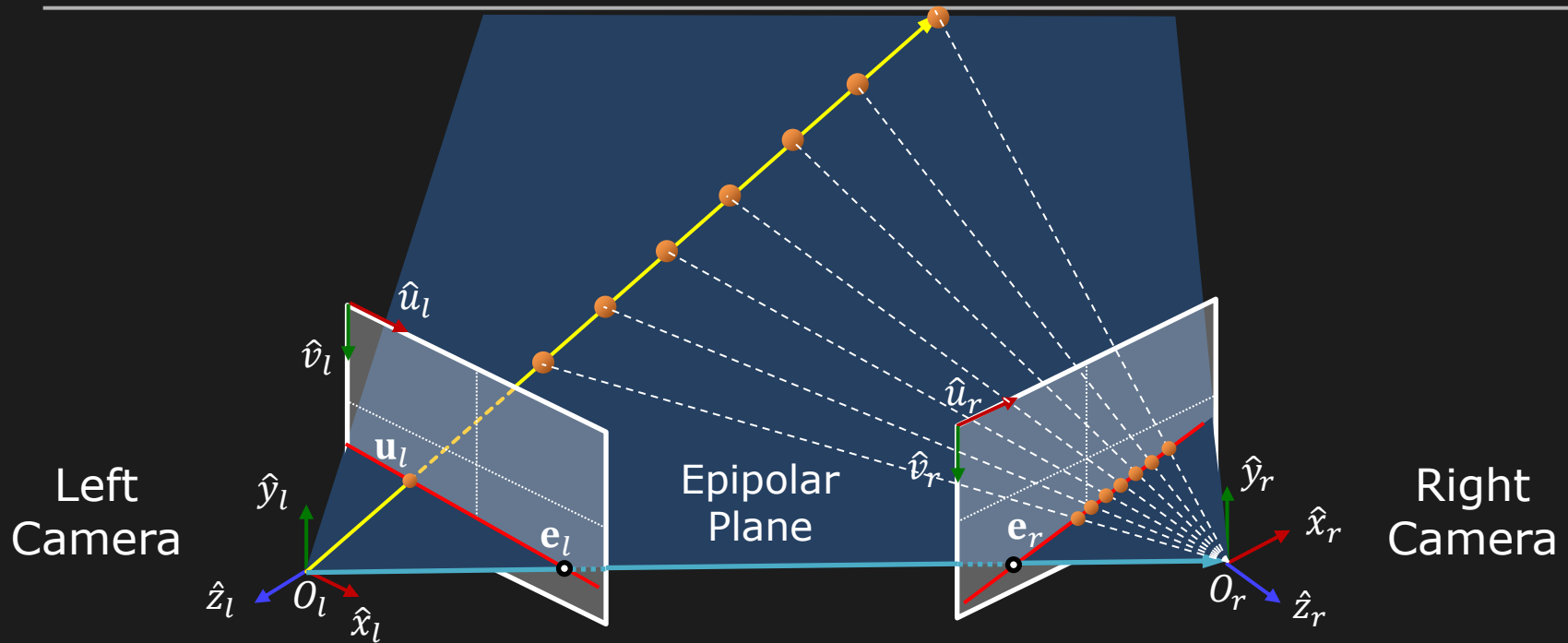
Epipolar Geometry: Epipolar Line



Intersection of the image plane and epipolar plane is the **Epipolar Line**.

Each scene point corresponds to **two Epipolar Lines**, one each on the two image planes.

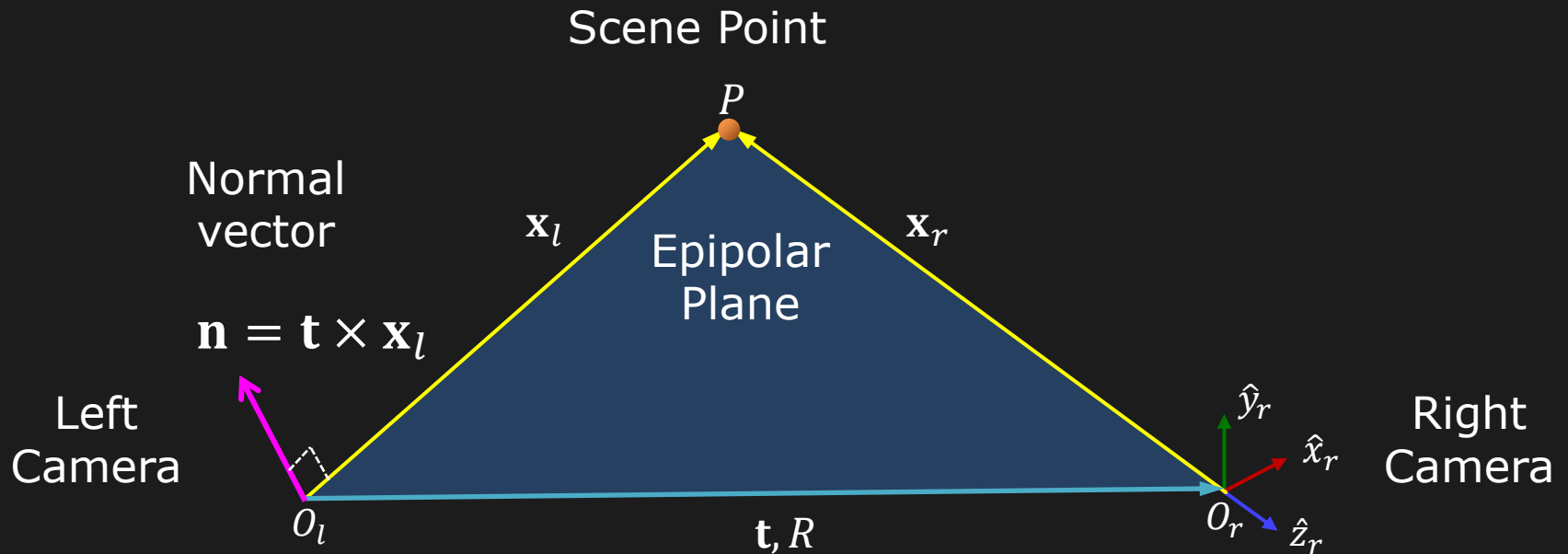
Epipolar Geometry: Epipolar Constraint



Given a point in one image, the corresponding point in the other image must lie on the epipolar line.

Epipolar constraint reduces the problem of finding correspondence to a 1D search.

Epipolar Constraint



Vector normal to the epipolar plane: $\mathbf{n} = \mathbf{t} \times \mathbf{x}_l$

Dot product of $\mathbf{n}$ and $\mathbf{x}_l$ (perpendicular vectors) is zero.

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$$

Epipolar Constraint in Matrix Form

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$$

$$[x_l \quad y_l \quad z_l] \begin{bmatrix} t_y z_l - t_z y_l \\ t_z x_l - t_x z_l \\ t_x y_l - t_y x_l \end{bmatrix} = 0 \quad \text{Cross-product definition}$$

$$[x_l \quad y_l \quad z_l] \underbrace{\begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}}_{T_{\times}} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = 0 \quad \text{Matrix-vector form}$$

But we know that:

$$\begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

The Epipolar Constraint

Substituting into the epipolar constraint gives:

$$[x_l \quad y_l \quad z_l] \left(\begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} + \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \right) = 0$$

$t \times t = 0$


$$[x_l \quad y_l \quad z_l] \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Essential Matrix E

$$E = T_{\times} R$$

The Essential Matrix E

Essential Matrix E : Relates position of scene point in left camera coordinate (x_l, y_l, z_l) to position in right camera coordinates (x_r, y_r, z_r)

$$\mathbf{x}_l \cdot E \mathbf{x}_r = 0$$

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

3D position in left
camera coordinates

3x3 Essential
Matrix

3D position in right
camera coordinates

Epipolar Constraint in Image Coordinates

Forward imaging equations:

$$\begin{bmatrix} \tilde{u}_l \\ \tilde{v}_l \\ \tilde{w}_l \end{bmatrix} = \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} \\ 0 & f_y^{(l)} & o_y^{(l)} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix}$$

$$\tilde{\mathbf{u}}_l = \mathbf{K}_l \mathbf{x}_l$$

$$\begin{bmatrix} \tilde{u}_r \\ \tilde{v}_r \\ \tilde{w}_r \end{bmatrix} = \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} \\ 0 & f_y^{(r)} & o_y^{(r)} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix}$$

$$\tilde{\mathbf{u}}_r = \mathbf{K}_r \mathbf{x}_r$$

Inverse imaging equations:

$$\begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix} = \begin{bmatrix} \frac{1}{f_x^{(l)}} & 0 & -\frac{o_x^{(l)}}{f_x^{(l)}} \\ 0 & \frac{1}{f_y^{(l)}} & -\frac{o_y^{(l)}}{f_y^{(l)}} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tilde{u}_l \\ \tilde{v}_l \\ \tilde{w}_l \end{bmatrix}$$

$$\mathbf{x}_l = \mathbf{K}_l^{-1} \tilde{\mathbf{u}}_l$$

$$\begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} \frac{1}{f_x^{(r)}} & 0 & -\frac{o_x^{(r)}}{f_x^{(r)}} \\ 0 & \frac{1}{f_y^{(r)}} & -\frac{o_y^{(r)}}{f_y^{(r)}} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tilde{u}_r \\ \tilde{v}_r \\ \tilde{w}_r \end{bmatrix}$$

$$\mathbf{x}_r = \mathbf{K}_r^{-1} \tilde{\mathbf{u}}_r$$

Epipolar Constraint in Image Coordinates

Rewriting the epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Substituting with the inverse imaging equations gives:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \underbrace{\begin{bmatrix} \frac{1}{f_x^{(l)}} & 0 & 0 \\ -\frac{o_x^{(l)}}{f_x^{(l)}} & \frac{1}{f_y^{(l)}} & 0 \\ 0 & -\frac{o_y^{(l)}}{f_y^{(l)}} & 1 \end{bmatrix}}_{(K_l^{-1})^T} \underbrace{\begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix}}_E \underbrace{\begin{bmatrix} \frac{1}{f_x^{(r)}} & 0 & -\frac{o_x^{(r)}}{f_x^{(r)}} \\ 0 & \frac{1}{f_y^{(r)}} & -\frac{o_y^{(r)}}{f_y^{(r)}} \\ 0 & 0 & 1 \end{bmatrix}}_{K_r^{-1}} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Epipolar Constraint in Image Coordinates

Rewriting the epipolar constraint:

$$\begin{bmatrix} x_l & y_l & z_l \end{bmatrix} \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Substituting with the inverse imaging equations gives:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix F

$$F = (K_l^{-1})^T E K_r^{-1}$$

The Fundamental Matrix F

Fundamental Matrix F : Relates position of scene point in left image $(u_l, v_l, 1)$ to position of the same scene point in the right image $(u_r, v_r, 1)$

$$\tilde{\mathbf{u}}_l \cdot F \tilde{\mathbf{u}}_r = 0$$

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

Homogeneous 2D
vector in left image
coordinates

3x3 Fundamental
Matrix

Homogeneous 2D
vector in right image
coordinates

Scale of Fundamental Matrix F

Fundamental matrix acts on homogenous coordinates.

We know that:

$$\begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} \equiv k \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{w} \end{bmatrix} \quad (k \neq 0 \text{ is any constant})$$

That is:

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = \begin{bmatrix} u_l & v_l & 1 \end{bmatrix} k \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}$$

Therefore, Fundamental Matrices F and kF produce the same epipolar constraint.

Fundamental Matrix F needs to be determined only up to a scale factor.

Epipolar Lines

If we know the Fundamental matrix F then,

given a point (u_l, v_l) in the left image, we can find the line in the right image that the corresponding point must lie on,

and, given a point (u_r, v_r) in the right image, we can find the line in the left image that the corresponding point must lie on.

Epipolar Lines from F Matrix

Given F and (u_r, v_r) , the Epipolar Constraint Equation:

$$\underbrace{\begin{bmatrix} u_l & v_l & 1 \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}}_{\text{Known}} = 0$$

We can expand the matrix equation as:

$$\underbrace{(f_{11}u_r + f_{12}v_r + f_{13})}_{a}u_l + \underbrace{(f_{21}u_r + f_{22}v_r + f_{23})}_{b}v_l + \underbrace{(f_{31}u_r + f_{32}v_r + f_{33})}_{c} = 0$$



$$au_l + bv_l + c = 0$$

Equation for left epipolar line
Computer Vision, SJTU, Wei Shen

Epipolar Lines from F Matrix

Given F and (u_l, v_l) , the Epipolar Constraint Equation:

$$\underbrace{\begin{bmatrix} u_l & v_l & 1 \end{bmatrix}}_{\text{Known}} \underbrace{\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}}_{\text{Unknown}} = 0$$

We can expand the matrix equation as:

$$\underbrace{(f_{11}u_l + f_{21}v_l + f_{31})}_{a'}u_r + \underbrace{(f_{12}u_l + f_{22}v_l + f_{32})}_{b'}v_r + \underbrace{(f_{13}u_l + f_{23}v_l + f_{33})}_{c'} = 0$$



$$a'u_r + b'v_r + c' = 0$$

Equation for right epipolar line

Fundamental Matrix Example

Given the Fundamental matrix,

$$F = \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix}$$

and the **left** image point

$$\tilde{u}_l = \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix}$$

Left Image



Right Image



The equation for the epipolar line in the **right** image is

$$[u_r \quad v_r \quad 1] \begin{bmatrix} -.003 & -.003 & 2.97 \\ -.028 & -.008 & 56.38 \\ 13.19 & -29.2 & -9999 \end{bmatrix} \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix Example

Given the Fundamental matrix,

$$F = \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix}$$

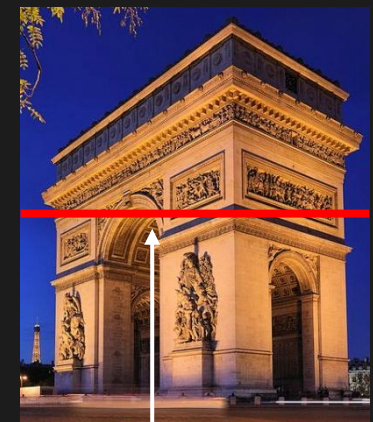
and the **left** image point

$$\tilde{u}_l = \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix}$$

Left Image



Right Image



Epipolar Line

The equation for the epipolar line in the **right** image is

$$.03u_r + .99v_r - 265 = 0$$

Fundamental Matrix Example

Given the Fundamental matrix,

$$F = \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix}$$

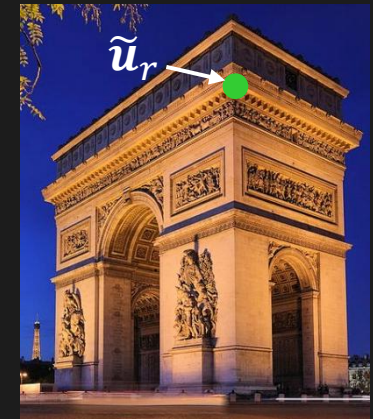
and the **right** image point

$$\tilde{u}_r = \begin{bmatrix} 205 \\ 80 \\ 1 \end{bmatrix}$$

Left Image



Right Image



The equation for the epipolar line in the **left** image is

$$\begin{bmatrix} u_l & v_l & 1 \end{bmatrix} \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix} \begin{bmatrix} 205 \\ 80 \\ 1 \end{bmatrix} = 0$$

Fundamental Matrix Example

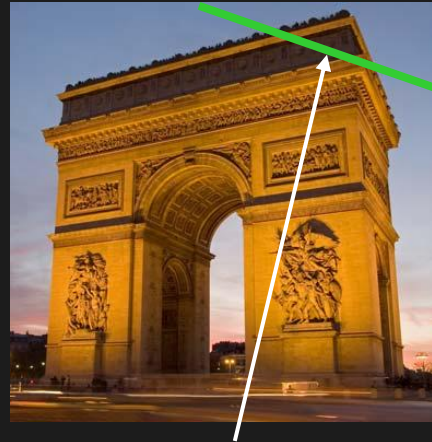
Given the Fundamental matrix,

$$F = \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix}$$

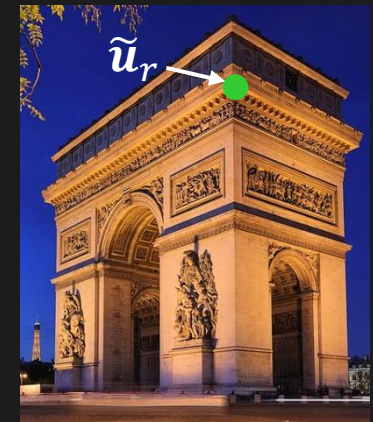
and the **right** image point

$$\tilde{u}_r = \begin{bmatrix} 205 \\ 80 \\ 1 \end{bmatrix}$$

Left Image



Right Image

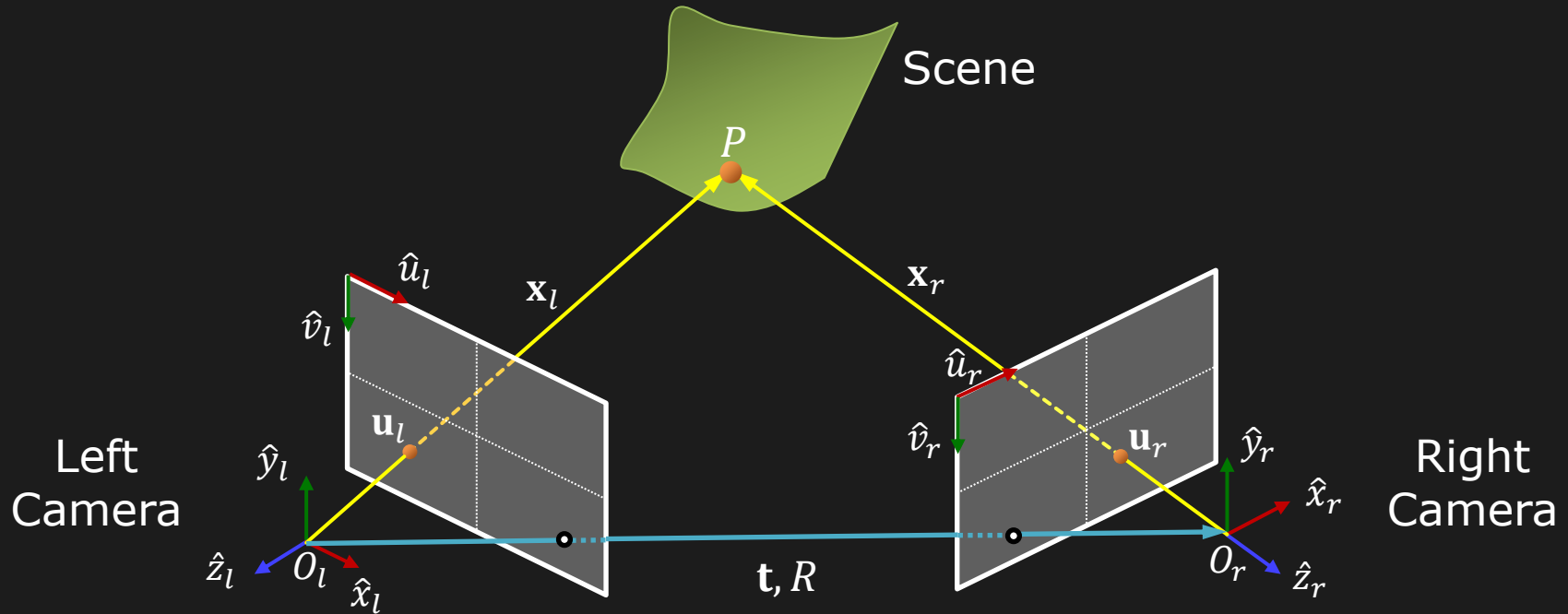


Epipolar Line

The equation for the epipolar line in the **left** image is

$$.32u_l - .95v_l - 151 = 0$$

Binocular Stereo



1. Assume Camera Intrinsic Parameters f_x, f_y, o_x, o_y are known.
2. Find Relative Camera Position t and Orientation R from the two images.
3. Find Correspondence for each pixel in the two images.
4. Compute Depth for each pixel using Triangulation.

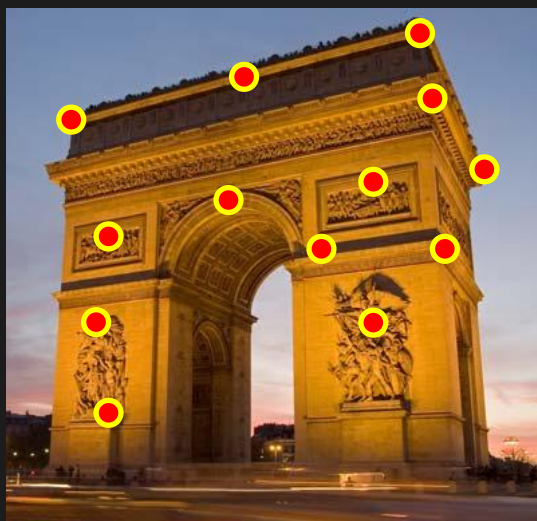
Stereo Calibration Using Fundamental Matrix

We use epipolar geometry to “Calibrate” the cameras to determine the relative camera position t and orientation R .

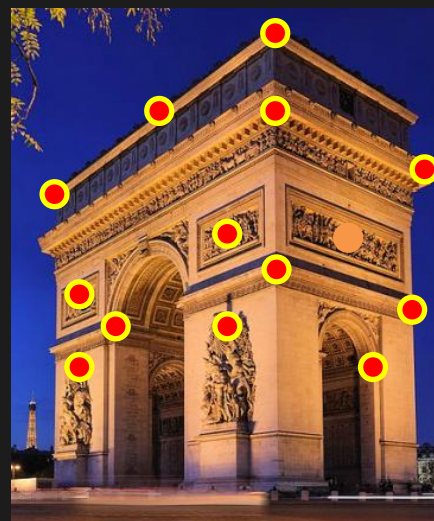
Stereo Calibration Procedure

Step 1: Find a set of features in left and right images (e.g. using SIFT)

Left image



Right image



Stereo Calibration Procedure

Step 2: Find correspondences by matching features.

Left image

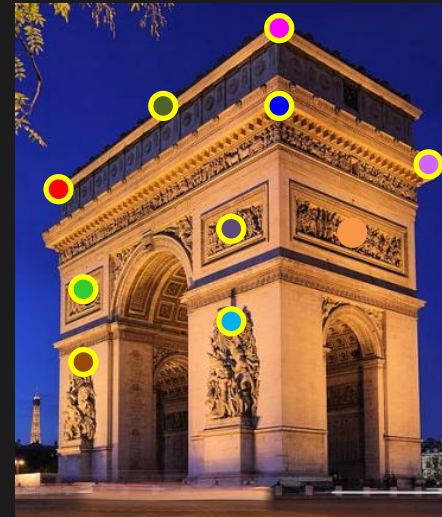


$$\bullet (u_l^{(1)}, v_l^{(1)})$$

$\vdots$

$$\bullet (u_l^{(m)}, v_l^{(m)})$$

Right image



$$\bullet (u_r^{(1)}, v_r^{(1)})$$

$\vdots$

$$\bullet (u_r^{(m)}, v_r^{(m)})$$

Stereo Calibration Procedure

Step 3: For each correspondence i , write out epipolar constraint

$$\underbrace{\begin{bmatrix} u_l^{(i)} & v_l^{(i)} & 1 \end{bmatrix}}_{\text{Known}} \underbrace{\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}}_{\text{Unknown}} \underbrace{\begin{bmatrix} u_r^{(i)} \\ v_r^{(i)} \\ 1 \end{bmatrix}}_{\text{Known}} = 0$$

Expand the matrix as linear equations

$$(f_{11}u_r^{(i)} + f_{12}v_r^{(i)} + f_{13})u_l^{(i)} + (f_{21}u_r^{(i)} + f_{22}v_r^{(i)} + f_{23})v_l^{(i)} + f_{31}u_r^{(i)} + f_{32}v_r^{(i)} + f_{33} = 0$$

Stereo Calibration Procedure

Rearranging the terms:

$$\begin{array}{c}
 \begin{bmatrix}
 u_l^{(1)} u_r^{(1)} & u_l^{(1)} v_r^{(1)} & u_l^{(1)} & v_l^{(1)} u_r^{(1)} & v_l^{(1)} v_r^{(1)} & v_l^{(1)} & u_r^{(1)} & v_r^{(1)} & 1 \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 u_l^{(i)} u_r^{(i)} & u_l^{(i)} v_r^{(i)} & u_l^{(i)} & v_l^{(i)} u_r^{(i)} & v_l^{(i)} v_r^{(i)} & v_l^{(i)} & u_l^{(i)} & u_r^{(i)} & 1 \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 u_l^{(m)} u_r^{(m)} & u_l^{(m)} v_r^{(m)} & u_l^{(m)} & v_l^{(m)} u_r^{(m)} & v_l^{(m)} v_r^{(m)} & v_l^{(m)} & u_l^{(m)} & u_r^{(m)} & 1
 \end{bmatrix}
 \begin{bmatrix}
 f_{11} \\
 f_{21} \\
 f_{31} \\
 f_{21} \\
 f_{22} \\
 f_{23} \\
 f_{31} \\
 f_{32} \\
 f_{33}
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0
 \end{bmatrix}
 \\
 \begin{array}{c}
 A \\
 \text{(Known)}
 \end{array}
 \qquad
 \begin{array}{c}
 \mathbf{f} \\
 \text{(Unknown)}
 \end{array}
 \end{array}$$

$$A \mathbf{f} = \mathbf{0}$$

Extracting Essential Matrix

Step 5: Given the intrinsic parameters of the two cameras, compute essential matrix E from the fundamental matrix F .

From definition:

$$F = (K_l^{-1})^T E K_r^{-1}$$

Therefore:

$$E = K_l^T F K_r$$

$$E = \begin{bmatrix} f_x^{(l)} & 0 & 0 \\ 0 & f_y^{(l)} & 0 \\ o_x^{(l)} & o_y^{(l)} & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} \\ 0 & f_y^{(r)} & o_y^{(r)} \\ 0 & 0 & 1 \end{bmatrix}$$

Extracting Rotation and Translation

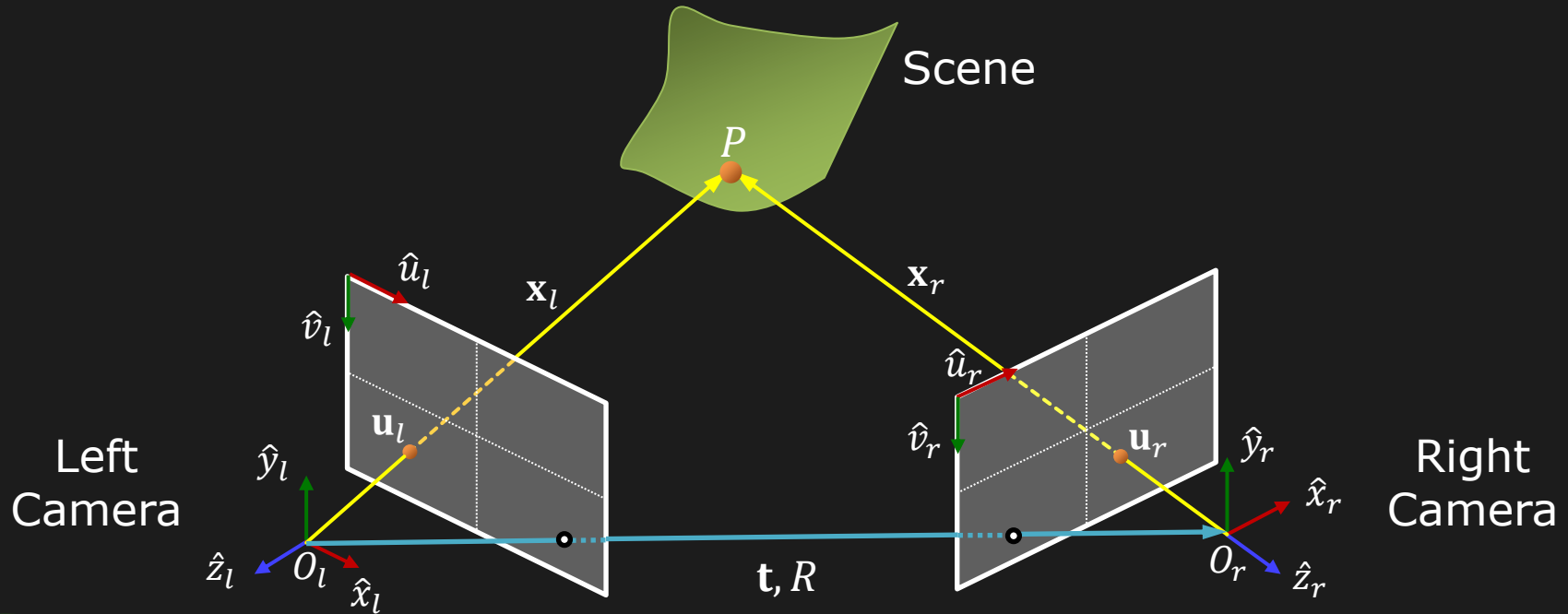
Step 6: Extract R and $\mathbf{t}$ from E

From definition, we know that:

$$E = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

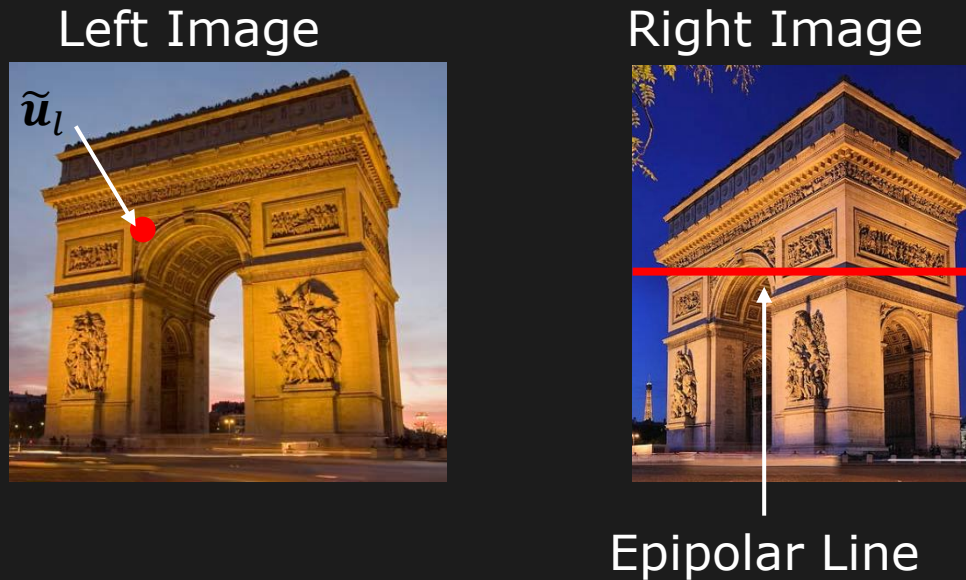
Given that $T_{\times}$ is a **skew-symmetric** matrix and R is an **orthonormal** matrix, it is possible to “decouple” $T_{\times}$ and R from their product using **SVD factorization** (see Appendix).

Binocular Stereo



- ✓ 1. Assume Camera Intrinsic Parameters f_x, f_y, o_x, o_y are known.
- ✓ 2. Find Relative Camera Position t and Orientation R from the two images.
3. Find Correspondence for each pixel in the two images.
4. Compute Depth for each pixel using Triangulation.

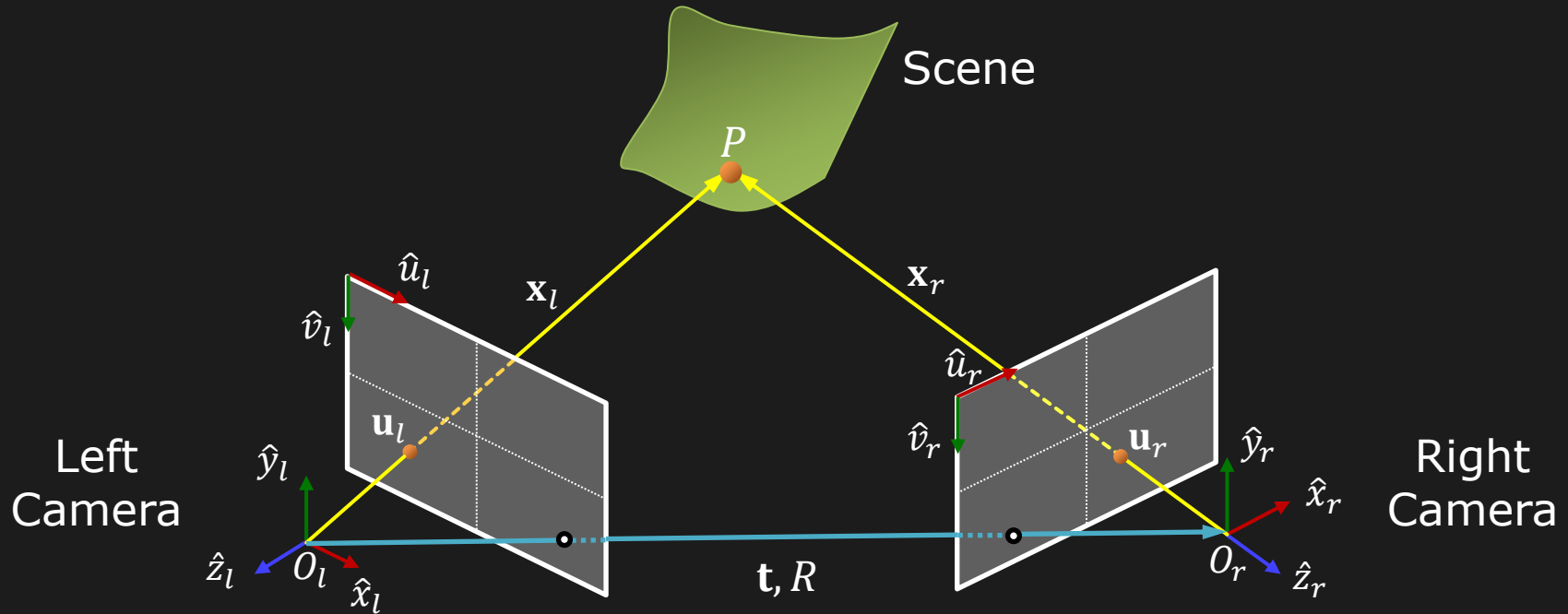
Correspondence using Fundamental Matrix



Given the Fundamental matrix and the **left** image point we can find the epipolar line in the right image and vice versa.

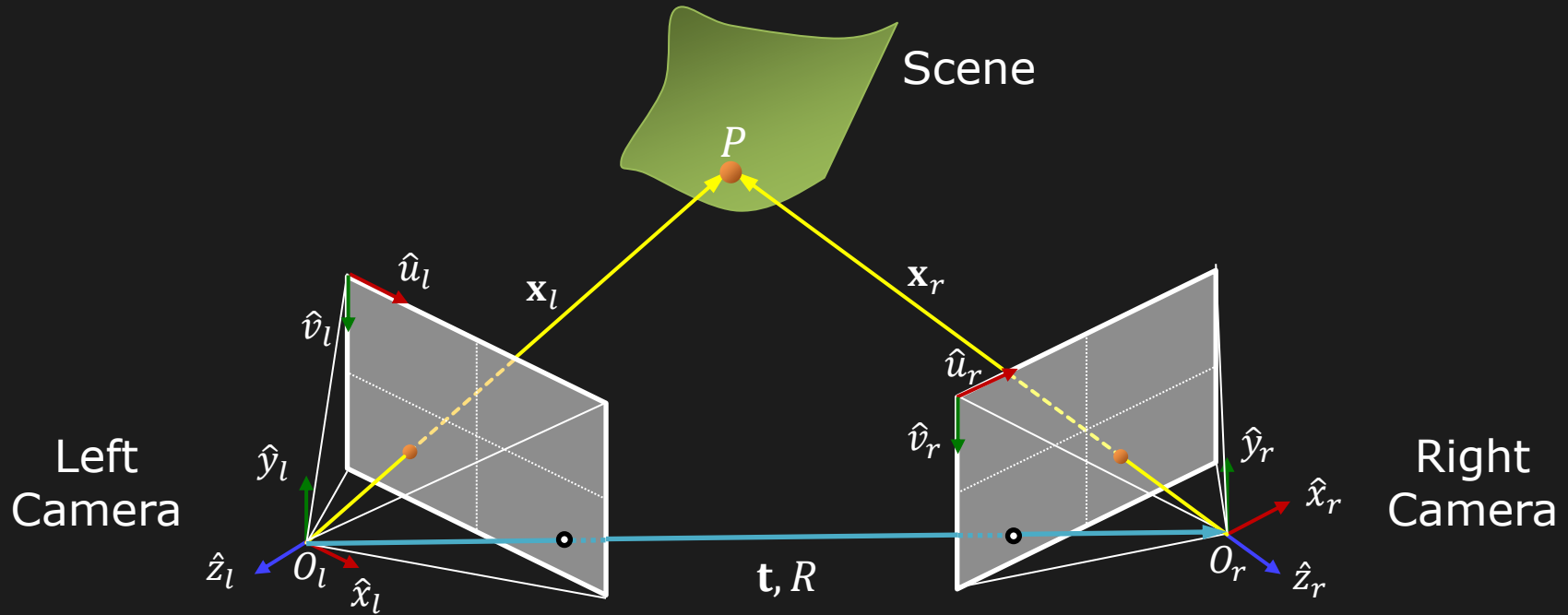
Perform template matching only along the epipolar line.
(See Simple Stereo/Image Processing I lectures)

Binocular Stereo



- ✓ 1. Assume Camera Intrinsic Parameters f_x, f_y, o_x, o_y are known.
- ✓ 2. Find Relative Camera Position t and Orientation R from the two images.
- ✓ 3. Find Correspondence for each pixel in the two images.
- 4. Compute Depth for each pixel using Triangulation.

Computing Depth



Given the intrinsic parameters, the projection of scene points on to the image sensor is given by:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix} \quad \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Computing Depth

Left Camera Imaging Equation

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix}$$

Right Camera Imaging Equation

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

We also know that relative position and orientation between the two cameras.

$$\begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Computing Depth

Left Camera Imaging Equation:

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(l)} & 0 & o_x^{(l)} & 0 \\ 0 & f_y^{(l)} & o_y^{(l)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}}_l = \mathbf{M}_l \tilde{\mathbf{x}}_r$$

Right Camera Imaging Equation:

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} f_x^{(r)} & 0 & o_x^{(r)} & 0 \\ 0 & f_y^{(r)} & o_y^{(r)} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

$$\tilde{\mathbf{u}}_r = \mathbf{P}_r \tilde{\mathbf{x}}_r$$

Computing Depth

Expanding the imaging equations:

$$\tilde{\mathbf{u}}_r = \mathbf{P}_r \tilde{\mathbf{x}}_r$$

$$\begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} \equiv \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Known

Unknown

$$\tilde{\mathbf{u}}_l = \mathbf{M}_l \tilde{\mathbf{x}}_r$$

$$\begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} \equiv \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

Known

Unknown

Rearranging the terms:

$$\begin{bmatrix} u_r p_{31} - p_{11} & u_r p_{32} - p_{12} & u_r p_{33} - p_{13} \\ v_r p_{31} - p_{21} & v_r p_{32} - p_{22} & v_r p_{33} - p_{23} \\ u_l m_{31} - m_{11} & u_l m_{32} - m_{12} & u_l m_{33} - m_{13} \\ v_l m_{31} - m_{21} & v_l m_{32} - m_{22} & v_l m_{33} - m_{23} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} p_{14} - p_{34} \\ p_{24} - p_{34} \\ m_{14} - m_{34} \\ m_{24} - m_{34} \end{bmatrix}$$

Computing Depth: Least Squares Solution

$$\begin{bmatrix} u_r p_{31} - p_{11} & u_r p_{32} - p_{12} & u_r p_{33} - p_{13} \\ v_r p_{31} - p_{21} & v_r p_{32} - p_{22} & v_r p_{33} - p_{23} \\ u_l m_{31} - m_{11} & u_l m_{32} - m_{12} & u_l m_{33} - m_{13} \\ v_l m_{31} - m_{21} & v_l m_{32} - m_{22} & v_l m_{33} - m_{23} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} p_{14} - p_{34} \\ p_{24} - p_{34} \\ m_{14} - m_{34} \\ m_{24} - m_{34} \end{bmatrix}$$

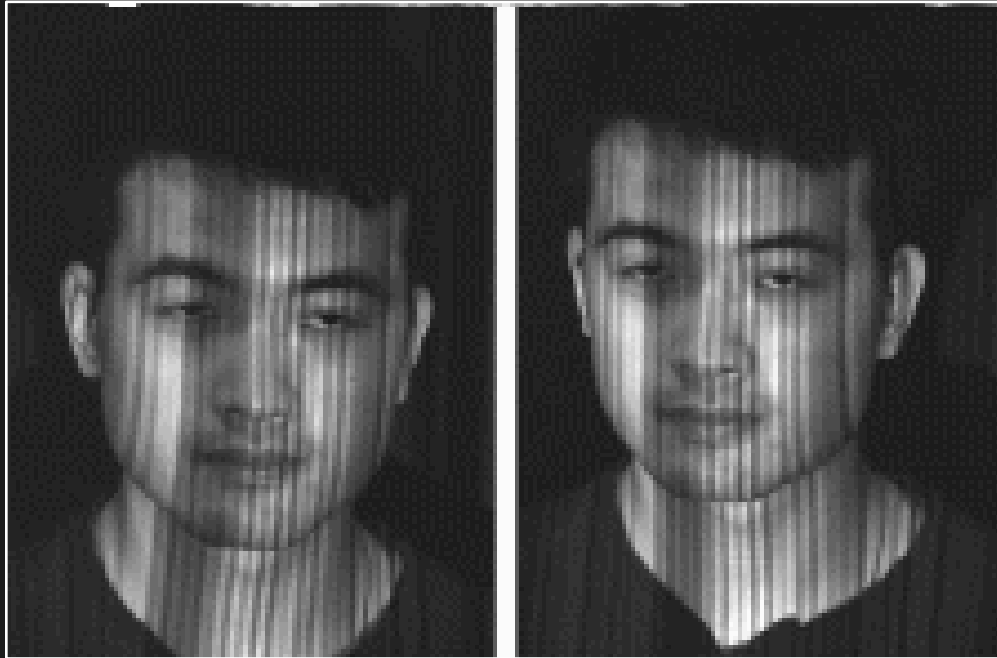
$A_{4 \times 3}$ $\mathbf{x}_r$ $\mathbf{b}_{4 \times 1}$
(Known) (Unknown) (Known)

Find least squares solution using pseudo-inverse:

$$\begin{aligned} A\mathbf{x}_r &= \mathbf{b} \\ A^T A\mathbf{x}_r &= A^T \mathbf{b} \end{aligned}$$

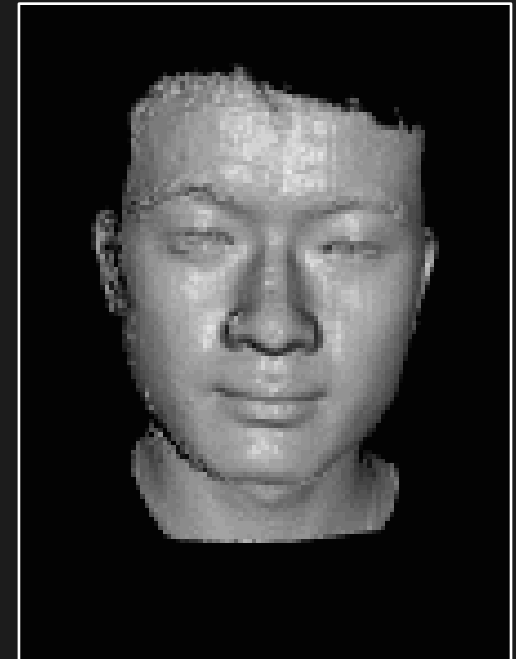
$$\mathbf{x}_r = (A^T A)^{-1} A^T \mathbf{b}$$

Results



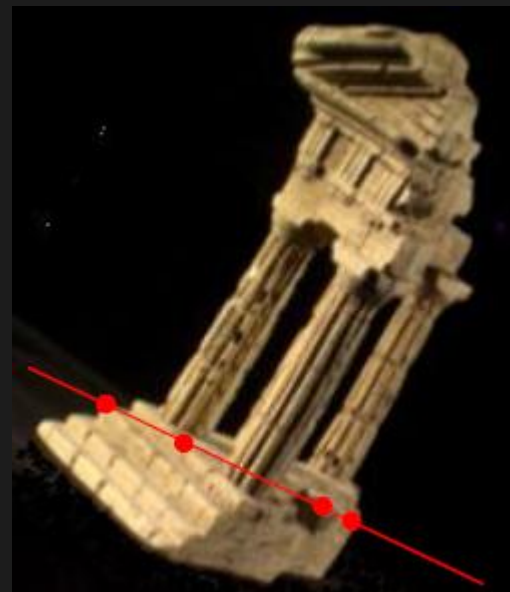
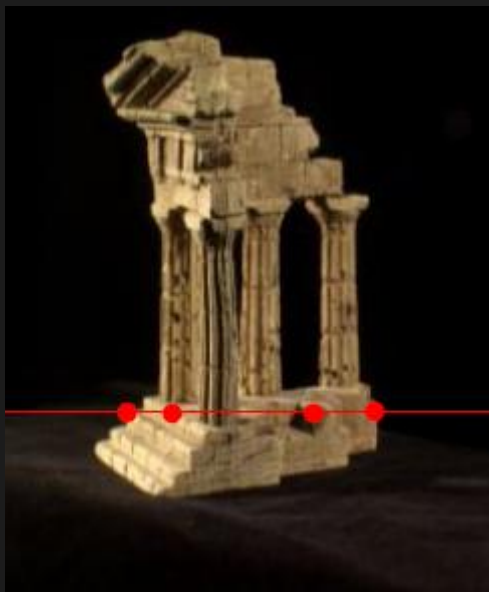
Left
Image

Right
Image



3D Structure

Reminder: Epipolar Geometry



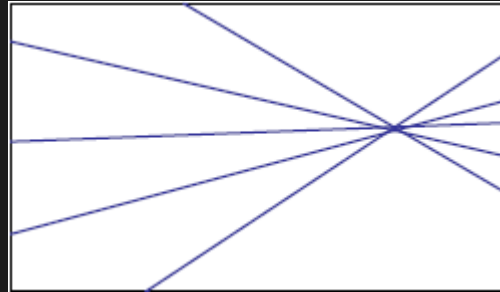
Since correspondences are constrained by epipolar geometry, only a one-dimensional search domain has to be considered.

Any problem here?

Stereo Rectification

In general: epipolar lines may be slanted.

=> Computation of the epipolar line for every image point is costly!



It is more desirable to have parallel, horizontal epipolar lines.

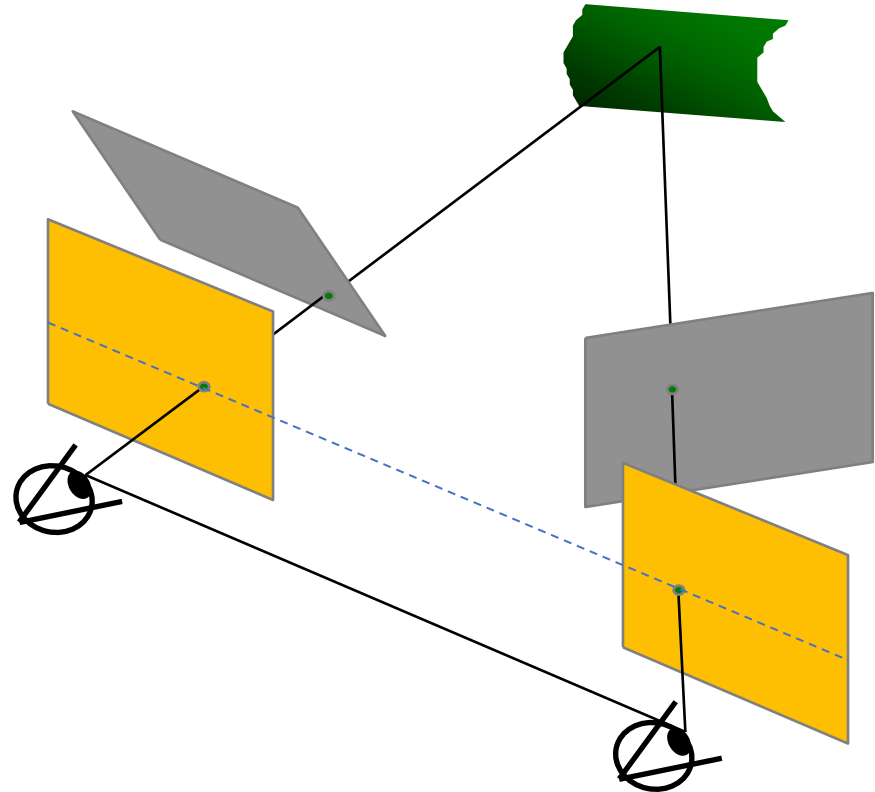
=> Simple search along scanlines!



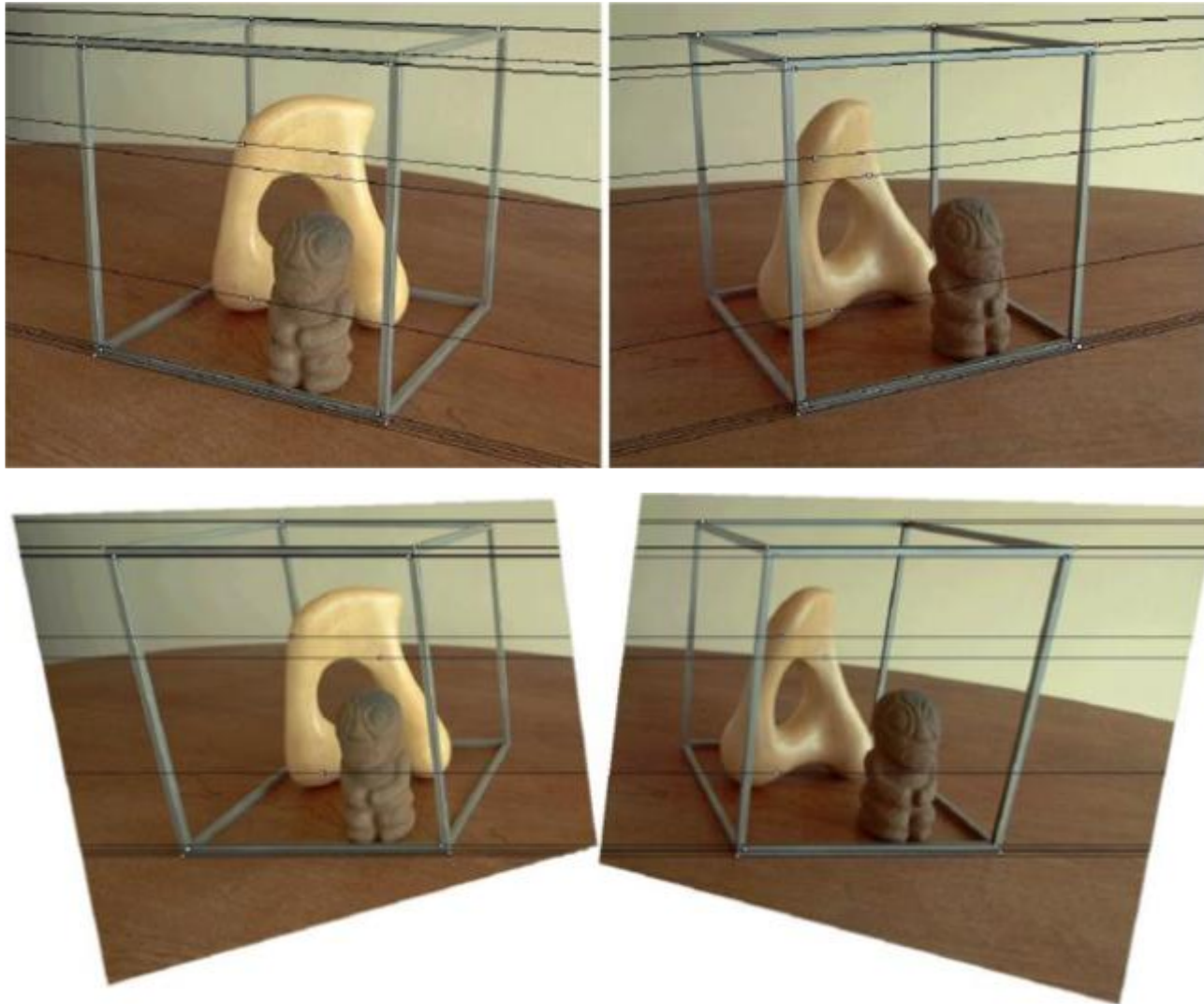
Stereo Rectification

Reproject image planes onto a common plane parallel to the line between camera centers

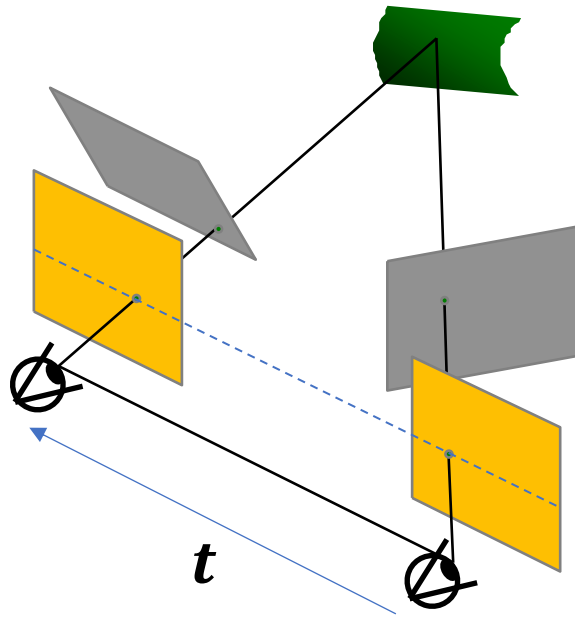
Need two homographies (3x3 transform), one for each input image reprojection



Stereo Rectification



How to make epipolar lines horizontal?



When this relationship holds:

$$\mathbf{R} = \mathbf{I} \quad \mathbf{t} = [T, 0, 0]^T$$

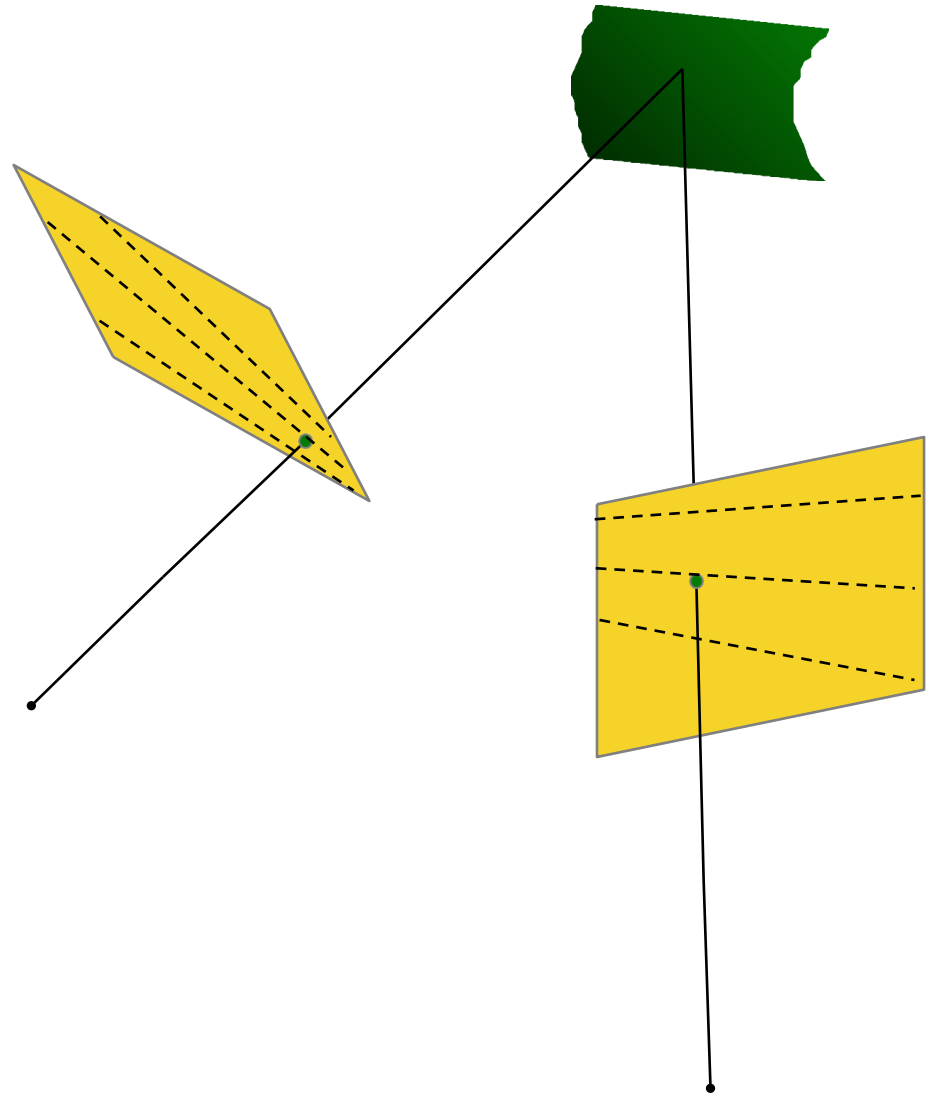
Make epipolar lines run parallel to the x-axis

By this convention, points will only move in the x-direction –there will be no movement in y-direction!

$$\left. \begin{aligned} \mathbf{E} = \mathbf{T}_\times \mathbf{R} &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -T \\ 0 & T & 0 \end{bmatrix} \\ [u_l \quad v_l \quad 1] \mathbf{K}^{-T} \mathbf{E} \mathbf{K}^{-1} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} &= 0 \end{aligned} \right\} \Rightarrow T v_l = T v_r$$

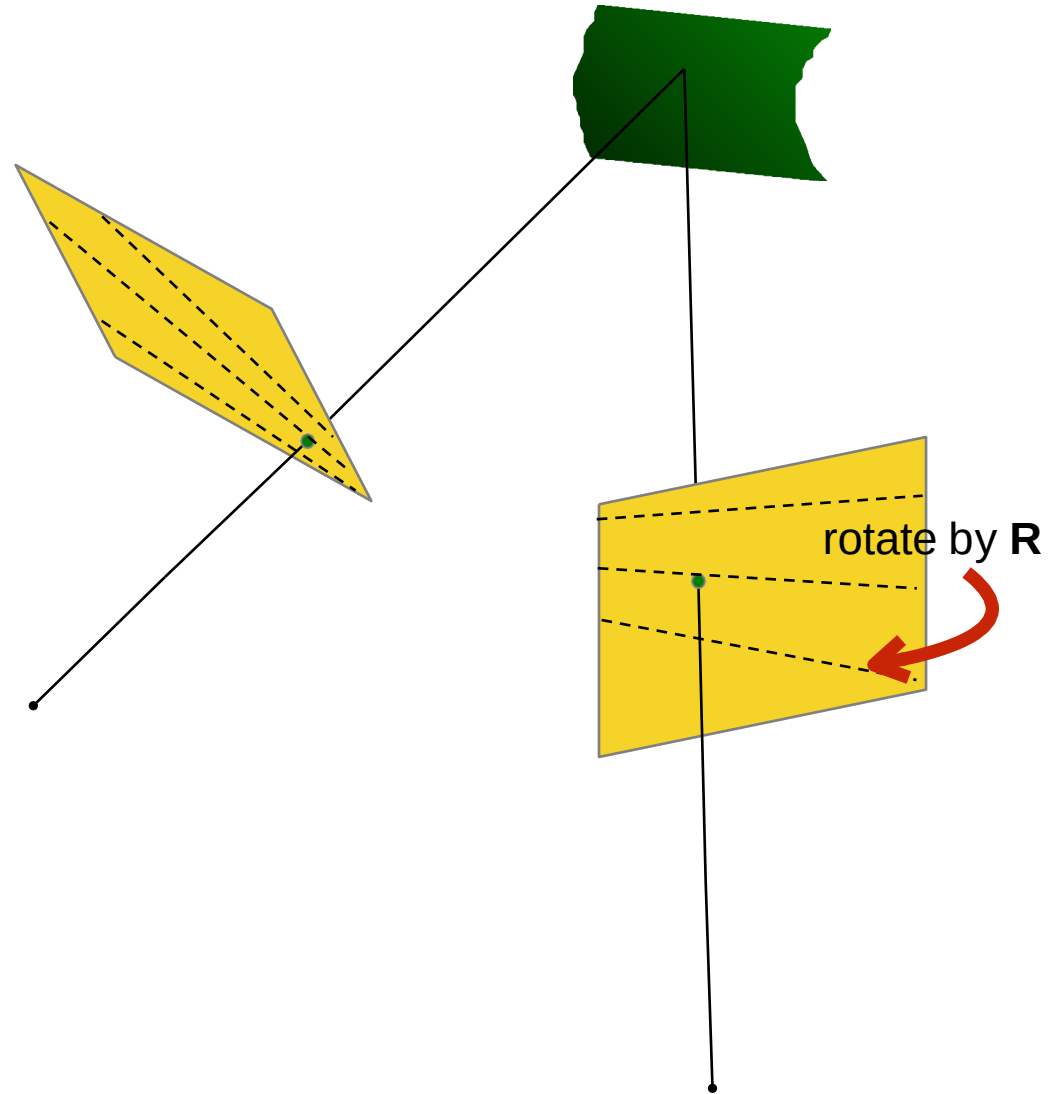
The image of a 3D point will always be on the same horizontal line

Stereo Rectification:



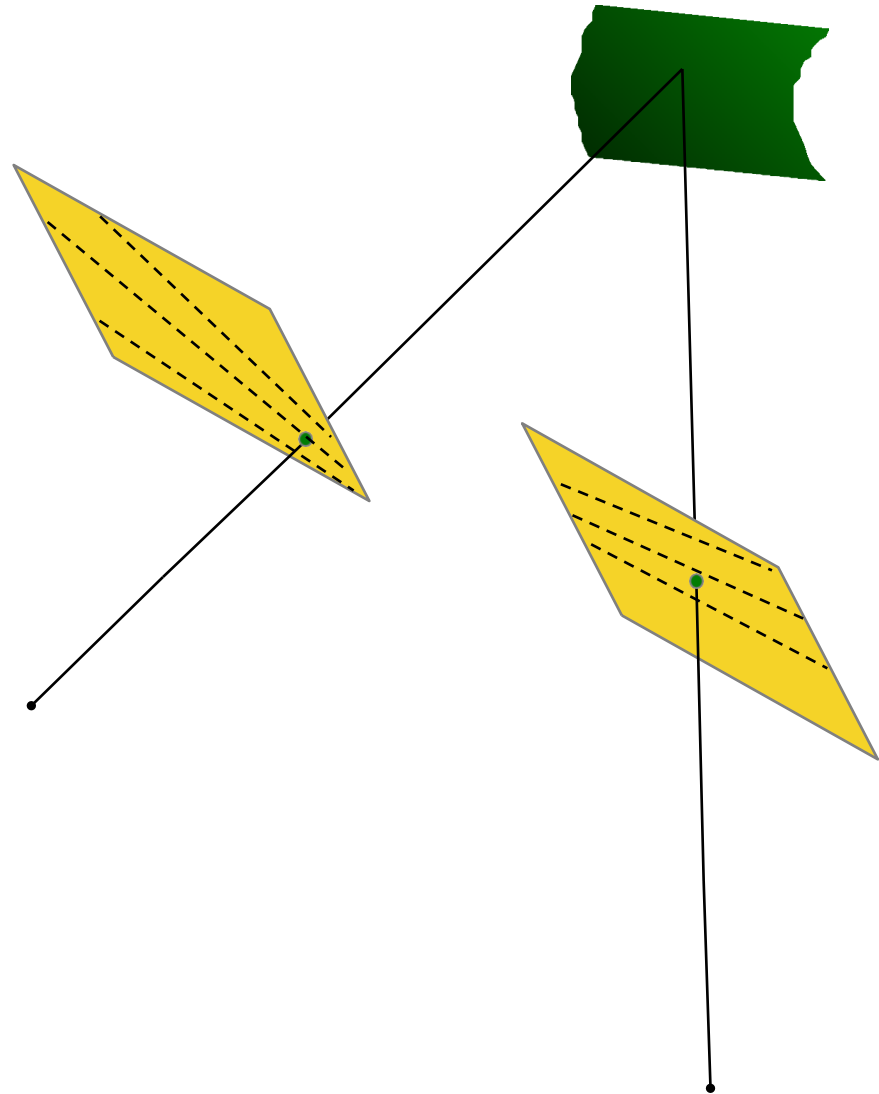
1. Compute $\mathbf{E}$ to get $\mathbf{R}$
2. Rotate right image by $\mathbf{R}$
3. Rotate both images by $\mathbf{R}_{\text{rect}}$
4. Scale both images by $\mathbf{H}$

Stereo Rectification:



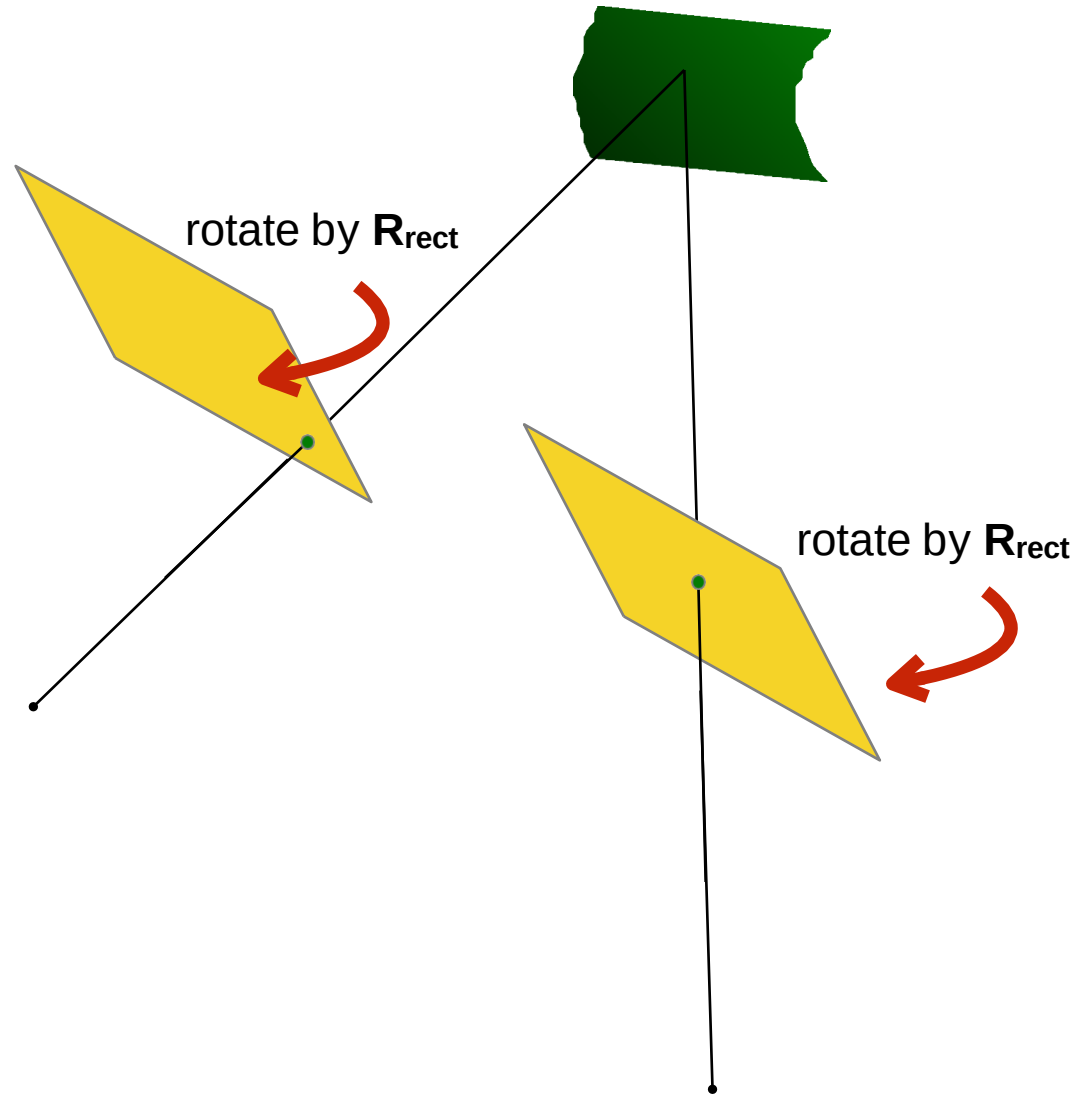
1. Compute E to get R
2. Rotate right image by R
3. Rotate both images by R_{rect}
4. Scale both images by H

Stereo Rectification:



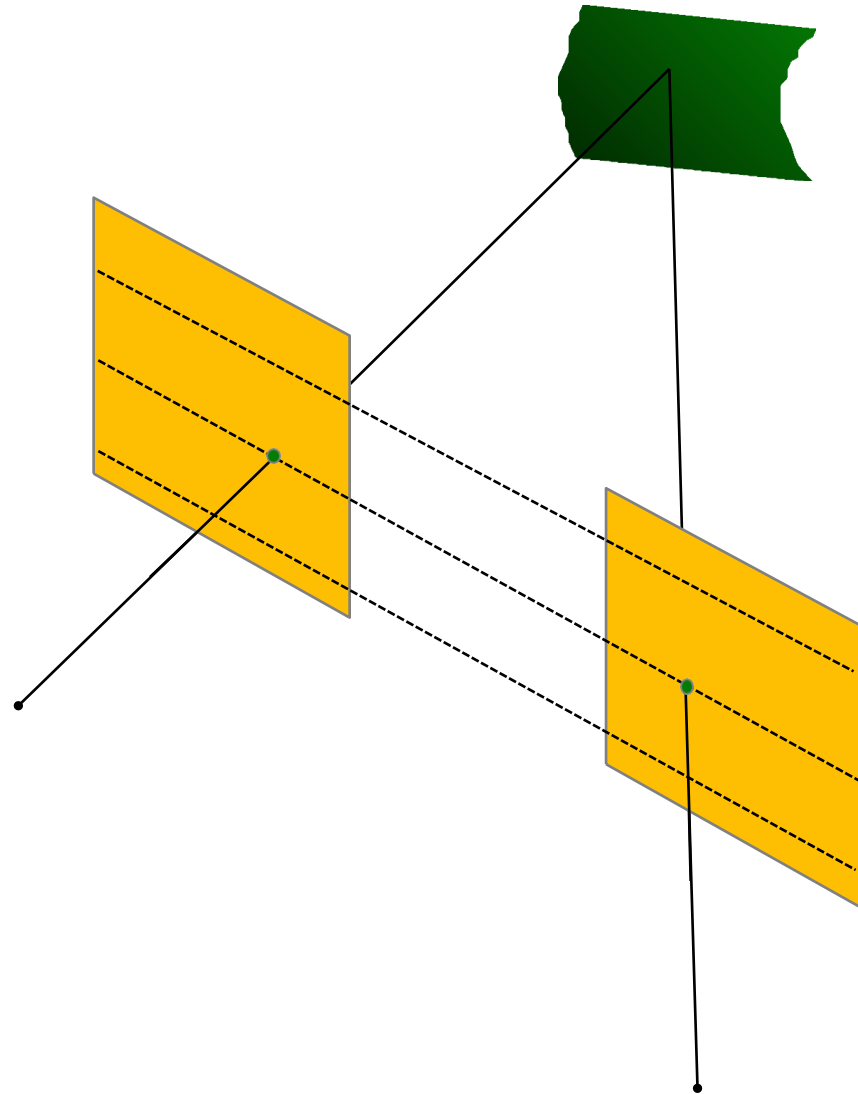
1. Compute $\mathbf{E}$ to get $\mathbf{R}$
2. Rotate right image by $\mathbf{R}$
3. Rotate both images by $\mathbf{R}_{\text{rect}}$
4. Scale both images by $\mathbf{H}$

Stereo Rectification:



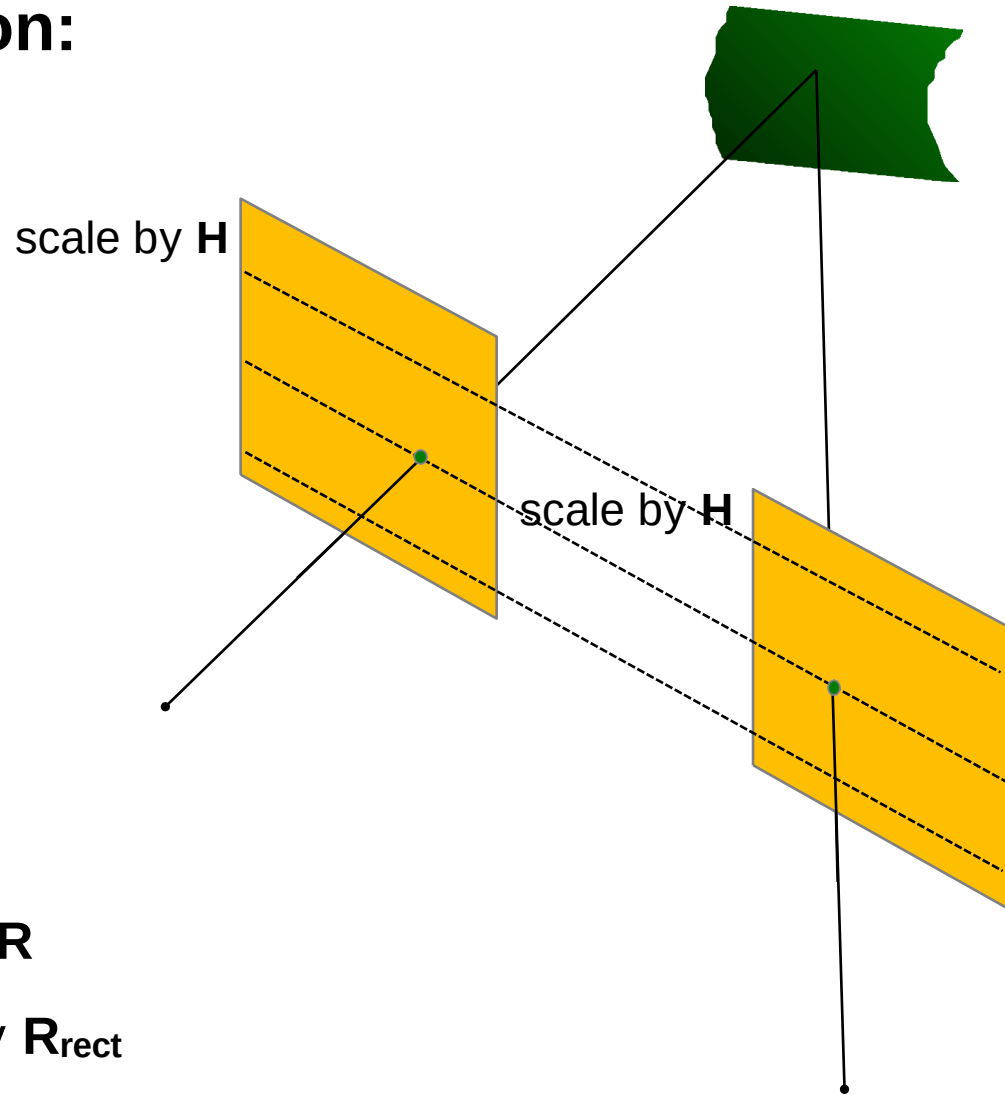
1. Compute $\mathbf{E}$ to get $\mathbf{R}$
2. Rotate right image by $\mathbf{R}$
3. Rotate both images by $\mathbf{R}_{\text{rect}}$
4. Scale both images by $\mathbf{H}$

Stereo Rectification:



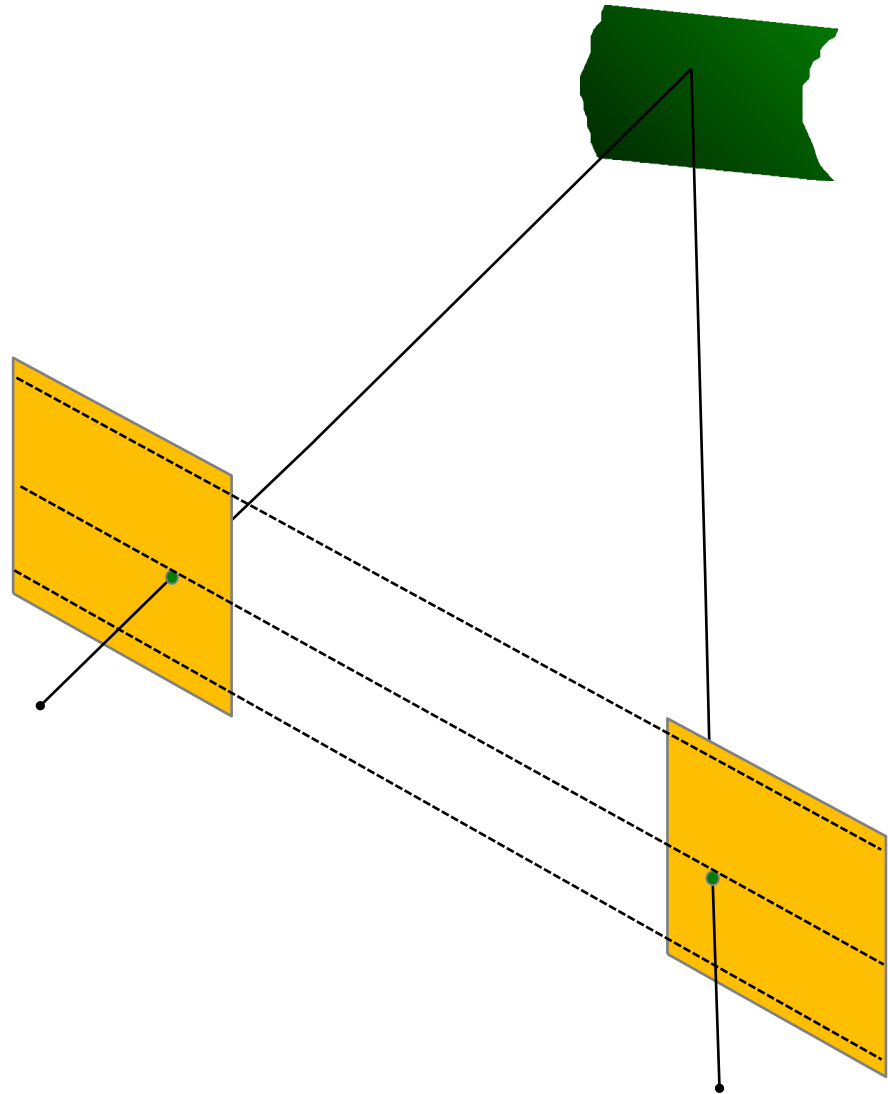
1. Compute $\mathbf{E}$ to get $\mathbf{R}$
2. Rotate right image by $\mathbf{R}$
3. Rotate both images by $\mathbf{R}_{\text{rect}}$
4. Scale both images by $\mathbf{H}$

Stereo Rectification:



1. Compute $\mathbf{E}$ to get $\mathbf{R}$
2. Rotate right image by $\mathbf{R}$
3. Rotate both images by $\mathbf{R}_{\text{rect}}$
4. Scale both images by H

Stereo Rectification:



1. Compute $\mathbf{E}$ to get $\mathbf{R}$
2. Rotate right image by $\mathbf{R}$
3. Rotate both images by $\mathbf{R}_{\text{rect}}$
4. Scale both images by $\mathbf{H}$

Building $\mathbf{R}_{\text{rect}}$ from $\mathbf{e}$

$$\text{Let } \mathbf{R}_{\text{rect}} = \begin{bmatrix} \mathbf{r}_1^\top \\ \mathbf{r}_2^\top \\ \mathbf{r}_3^\top \end{bmatrix}$$

Given:

epipole $\mathbf{e}$
(using SVD on $\mathbf{E}$)
(translation from $\mathbf{E}$)

$$\mathbf{r}_1 = \mathbf{e}_1 = \frac{\mathbf{t}}{|\mathbf{t}|}$$

epipole coincides with translation vector

$$\mathbf{r}_2 = \frac{\mathbf{t} \times [0 \ 0 \ 1]^\top}{|\mathbf{t} \times [0 \ 0 \ 1]^\top|} = \frac{1}{\sqrt{t_x^2 + t_y^2}} [-t_y \ t_x \ 0]^\top$$

cross product of $\mathbf{e}$ and
the direction vector of
the optical axis

$$\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2$$

orthogonal vector

Setting the epipole to where?

Since $\mathbf{r}_1 = \mathbf{e}_1 = \frac{\mathbf{t}}{|\mathbf{t}|}$ and orthogonal to $\mathbf{r}_2$ $\mathbf{r}_3$

Then

$$\mathbf{R}_{\text{rect}} \mathbf{e}_1 = \begin{bmatrix} \mathbf{r}_1^\top \mathbf{e}_1 \\ \mathbf{r}_2^\top \mathbf{e}_1 \\ \mathbf{r}_3^\top \mathbf{e}_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Where is this point located on the image plane?

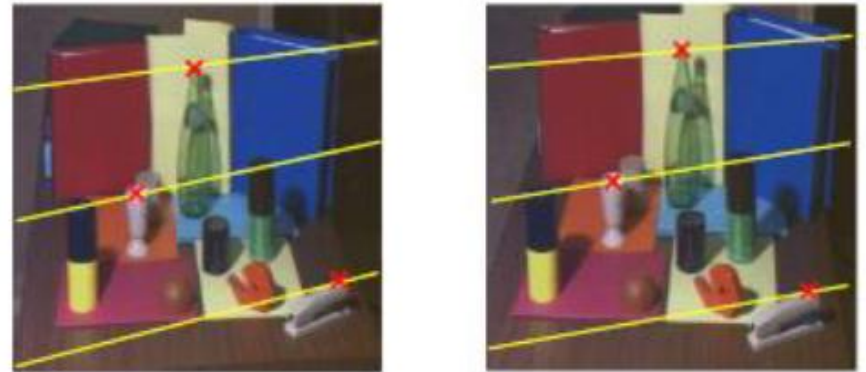
At x-infinity

Stereo Rectification Algorithm

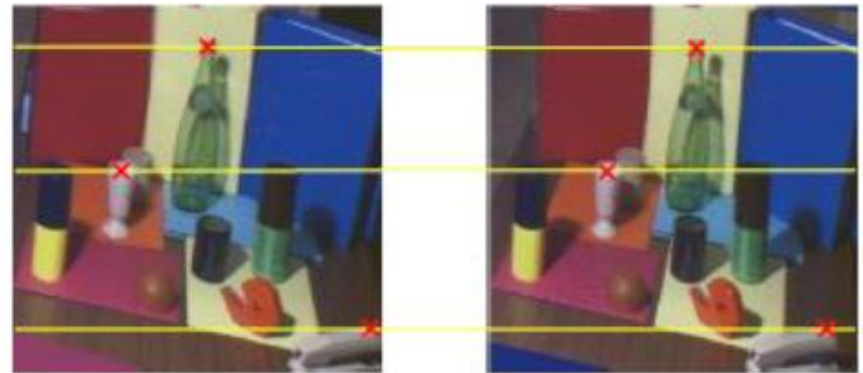
1. Estimate $\mathbf{E}$
2. Estimate the epipole $\mathbf{e}$ (SVD of $\mathbf{E}$)
3. Build $\mathbf{R}_{\text{rect}}$ from $\mathbf{e}$
4. Decompose $\mathbf{E}$ into $\mathbf{R}$ and $\mathbf{T}$
5. Set $\mathbf{R}_1 = \mathbf{R}_{\text{rect}}$ and $\mathbf{R}_2 = \mathbf{R}\mathbf{R}_{\text{rect}}$
6. Rotate each left camera point (warp image)
 $[x' \ y' \ z']^T = \mathbf{R}_1 [x \ y \ z]^T = \mathbf{R}_1 \mathbf{K}^{-1} \mathbf{p}'$
7. Rectified points as $\mathbf{p} = \mathbf{K}[x' \ y' \ z']^T$
8. Repeat 6 and 7 for right camera points using $\mathbf{R}_2$

Stereo Rectification Example

Verged, Unrectified



Non-Verged, Rectified

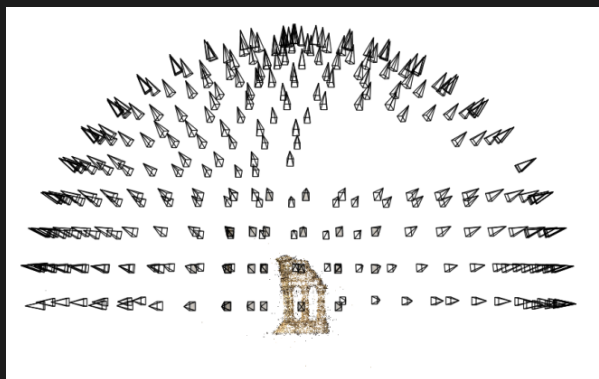


Structure from motion

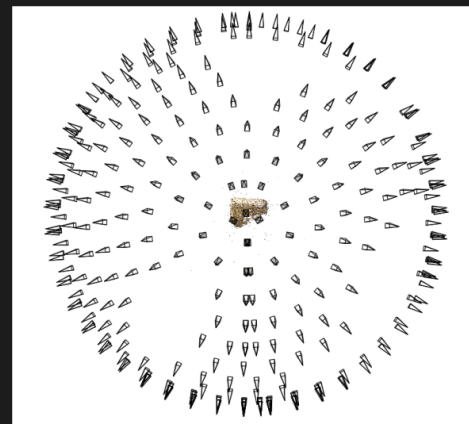
- What if I have lots of images?



Structure from motion



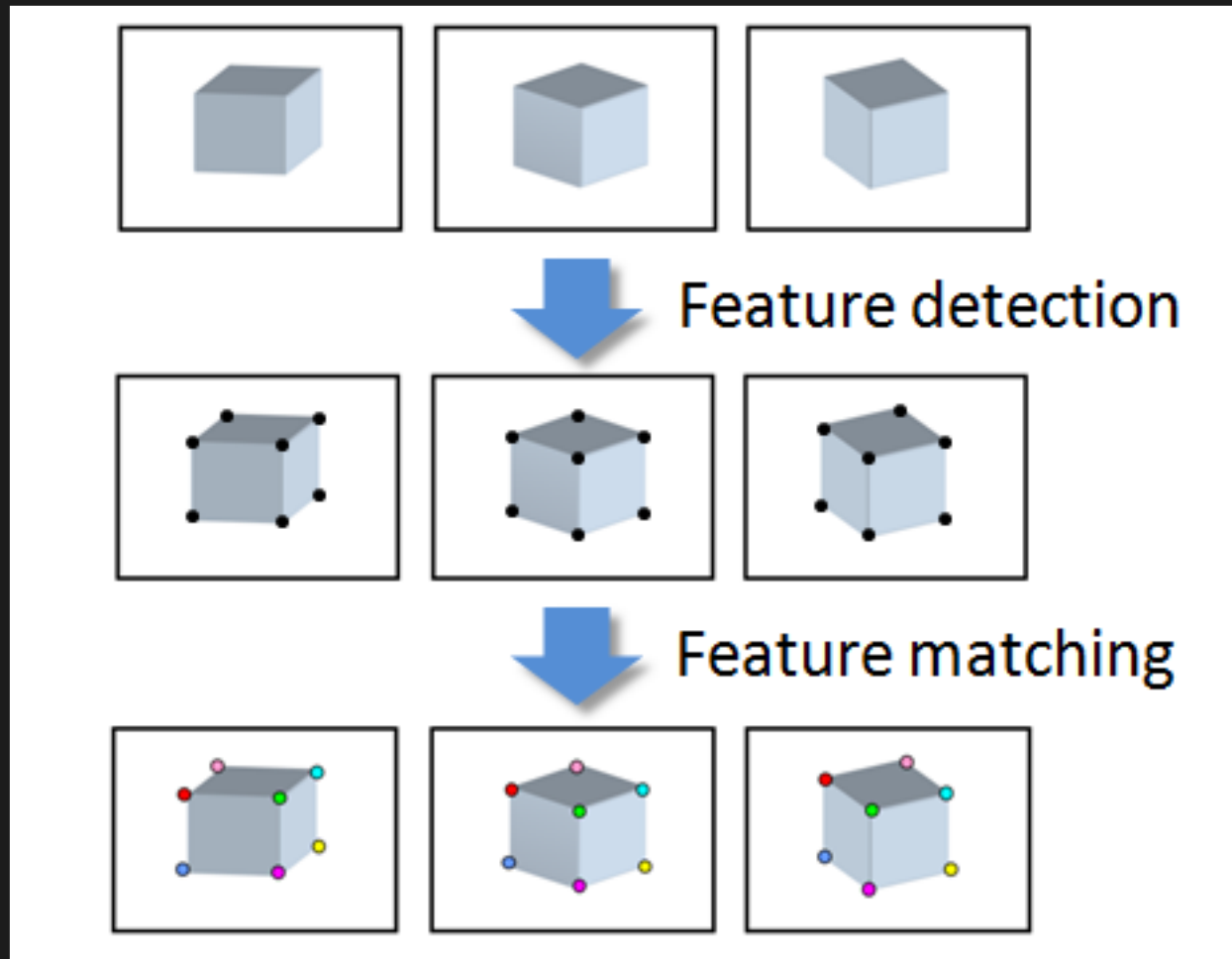
Reconstruction (side)



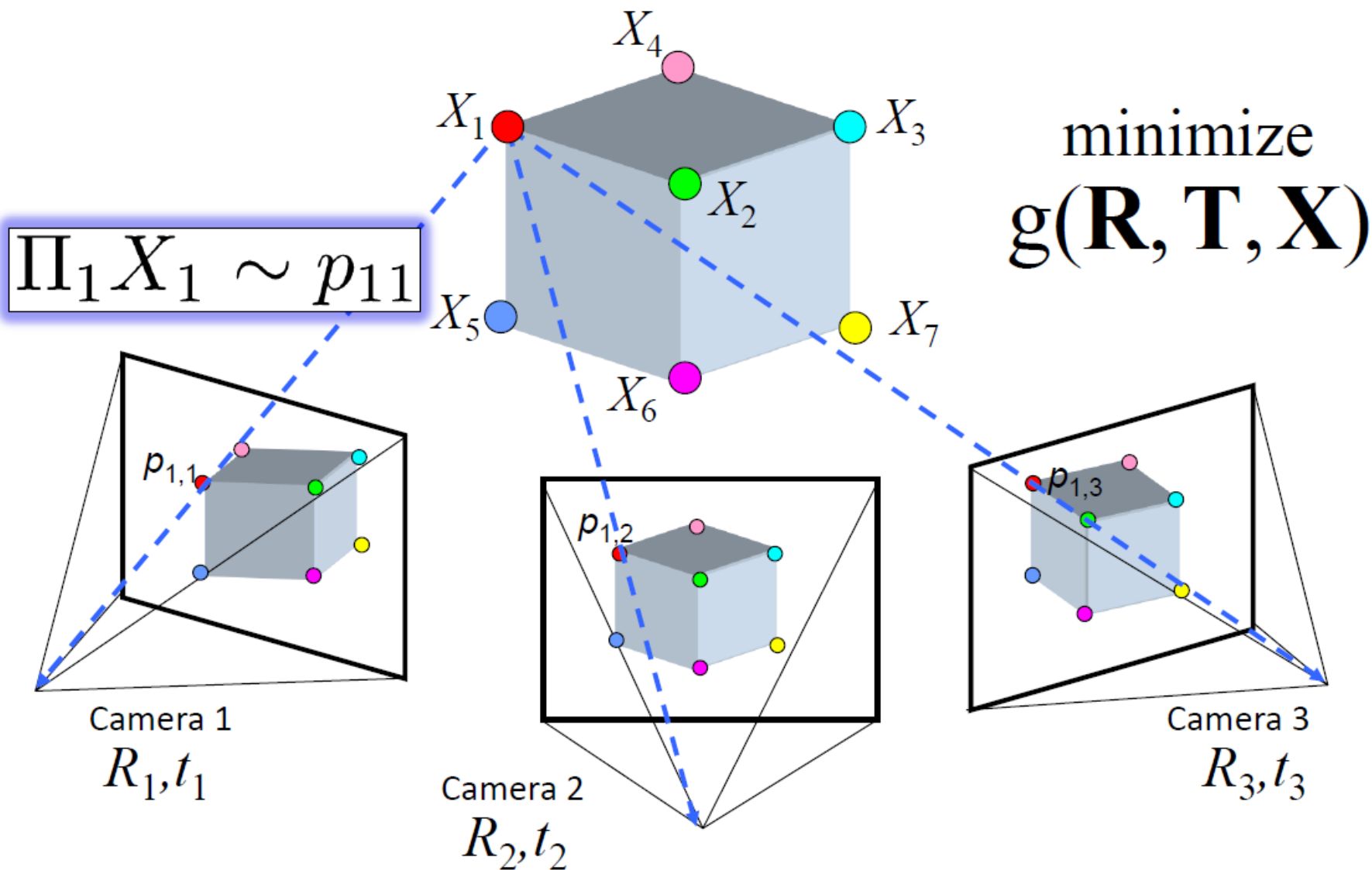
(top)

- Input: images with points in correspondence $\mathbf{p}_{i,j} = (u_{i,j}, v_{i,j})$
- Output
 - structure: 3D location $\mathbf{x}_i$ for each point $\mathbf{p}_i$
 - motion: camera parameters $\mathbf{R}_j$, $\mathbf{t}_j$ possibly $\mathbf{K}_i$
- Objective function: minimize reprojection error

Input



Structure from motion



Structure from motion

What are the variables (e.g., how do we represent “structure” and “motion”)?

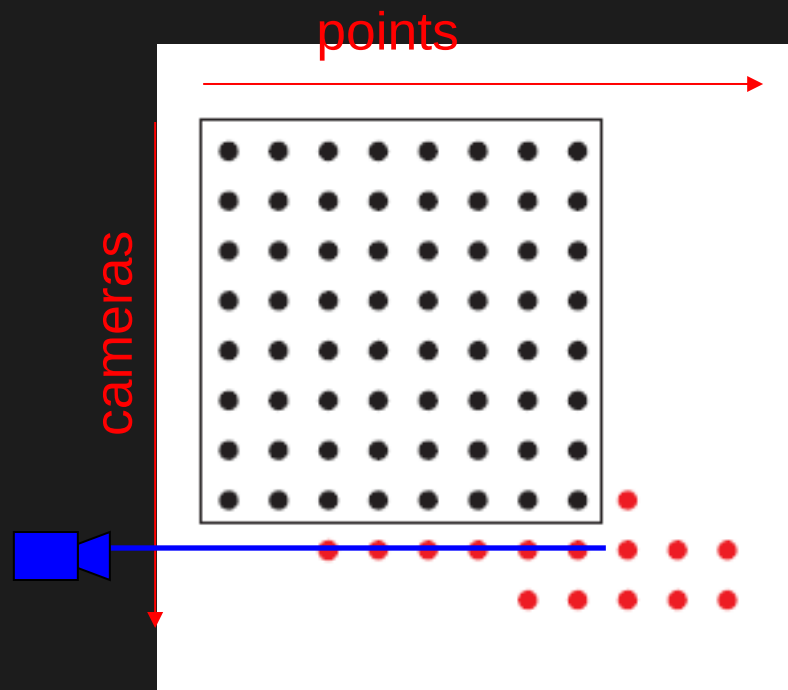
Minimize sum of squared reprojection errors:

$$g(\mathbf{X}, \mathbf{R}, \mathbf{T}) = \sum_{i=1}^m \sum_{j=1}^n \underbrace{w_{ij}}_{\substack{\text{indicator variable:} \\ \text{is point } i \text{ visible in image } j?}} \cdot \left\| \underbrace{\mathbf{P}(\mathbf{x}_i, \mathbf{R}_j, \mathbf{t}_j)}_{\substack{\text{predicted} \\ \text{image location}}} - \underbrace{\begin{bmatrix} u_{i,j} \\ v_{i,j} \end{bmatrix}}_{\substack{\text{observed} \\ \text{image location}}} \right\|^2$$

Initialize this using pairwise estimates of camera poses, compute depths, and then optimize jointly

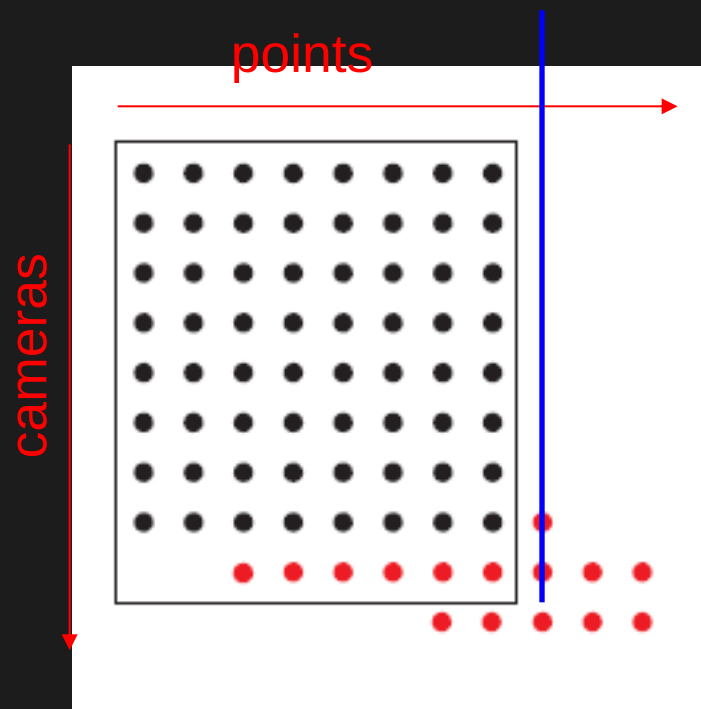
Sequential structure from motion

- Initialize motion from two images using fundamental matrix
- Initialize structure
- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*



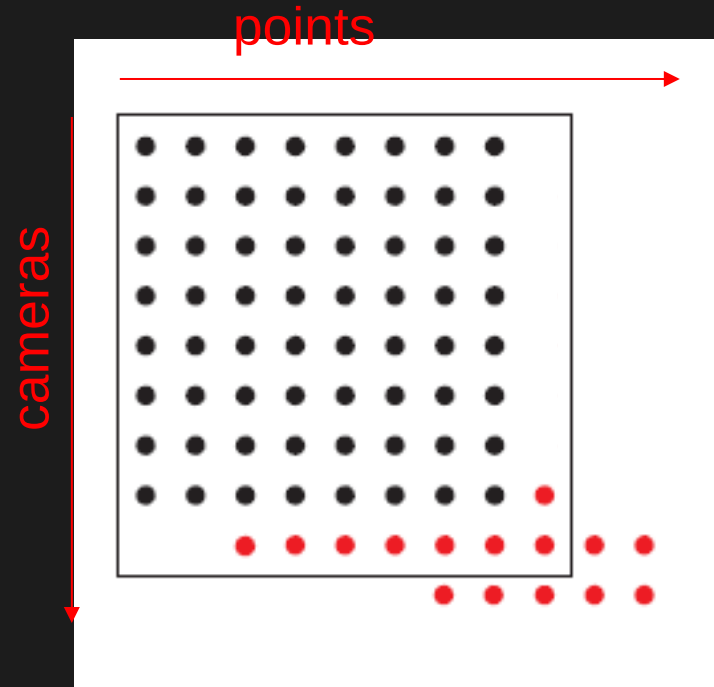
Sequential structure from motion

- Initialize motion from two images using fundamental matrix
- Initialize structure
- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*
 - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera – *triangulation*



Sequential structure from motion

- Initialize motion from two images using fundamental matrix
- Initialize structure
- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – *calibration*
 - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera – *triangulation*
- Refine structure and motion:
bundle adjustment - Optimized using non-linear least squares, e.g. Levenberg-Marquardt



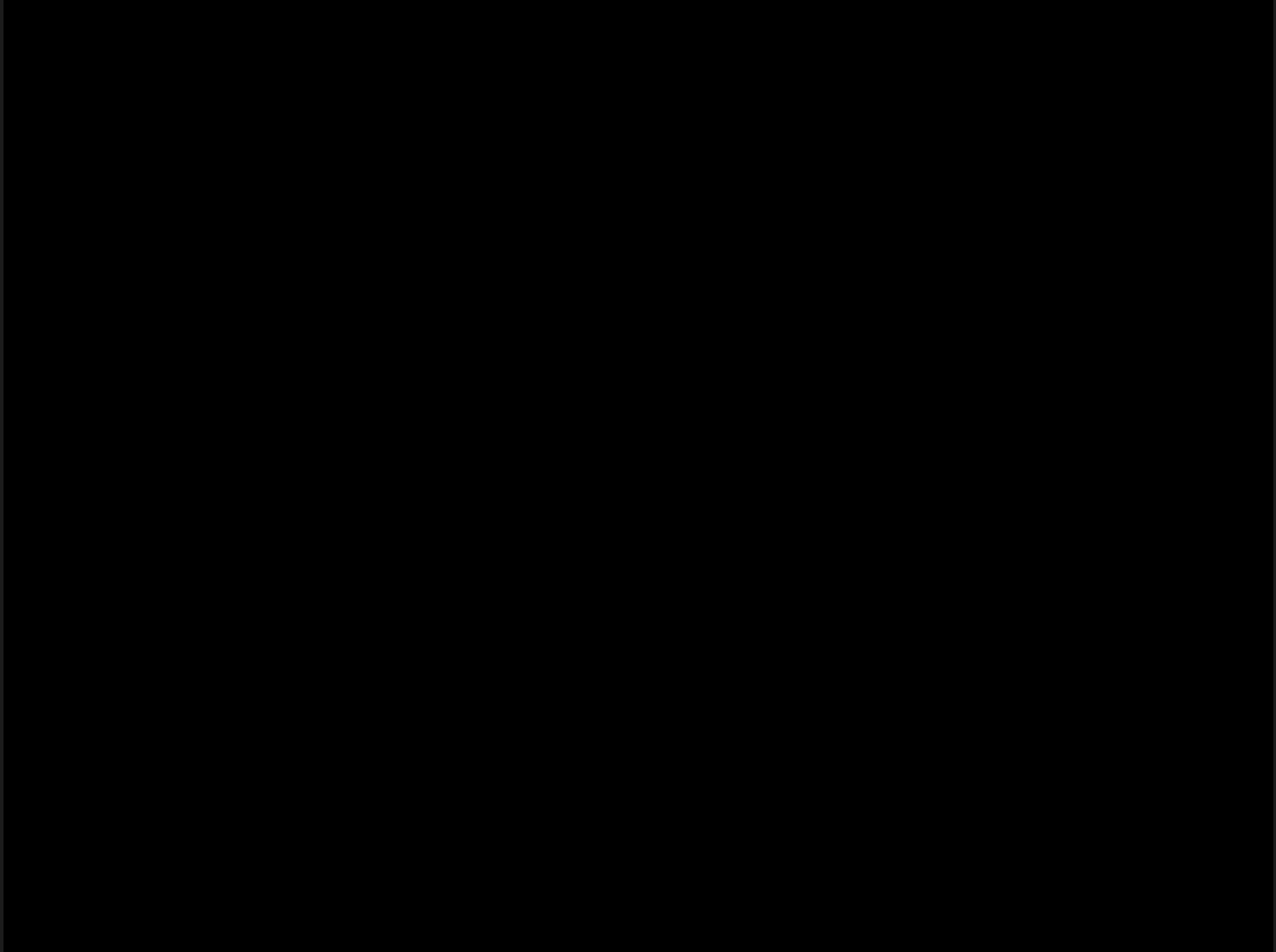
Results



3D Structure

Multiple views of the object

SFM Summed Up...



Appendix: Using the SVD to Extract R and $\mathbf{t}$

The SVD factorization of the Essential Matrix is

$$\begin{aligned} E &= U\Sigma V^T \\ &= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix}^T \end{aligned}$$

Where U and V are **orthonormal matrices**, and $(\sigma_1, \sigma_2, \sigma_3)$ are the **singular values** of the matrix E

The stereo calibration parameters R and $\mathbf{t}$ can be calculated from the SVD of E using the equations

$$R = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix}^T \quad \mathbf{t} = \begin{bmatrix} u_{13} \\ u_{23} \\ u_{33} \end{bmatrix}$$

References: Textbooks

Robot Vision (Chapter 13)

Horn, B. K. P., MIT Press

Computer Vision: A Modern Approach (Chapter 10)

Forsyth, D and Ponce, J., Prentice Hall

Multiple View Geometry (Chapters 8-10)

Hartley, R. and Zisserman, A., Cambridge University Press

Computer Vision: Algorithms and Applications (Chapter 7)

Szeliski, R., Springer

An introduction to 3D Computer Vision (Chapter 3)

Cyganek, B., Siebert, J. P., Wiley Pub

References: Papers

[Longuet-Higgins 1981] H.C. Longuet-Higgins. "A computer algorithm for reconstructing a scene from two projections." Nature, 1981.

[Fagueras 1992] O. Fagueras. "What can be seen in three dimensions with an uncalibrated stereo rig?." European Conference on Computer Vision, 1992.

Image Credits

- I.1 http://www.lasplash.com/publish/International_151/Hilton_Arc_de_Triomphe_Review-A_Parisian_Gem.php
- I.2 <http://nelietatravellingadventures.blogspot.com/2011/01/arc-de-triomphe-paris-france.html>
- I.3 <http://vision.middlebury.edu/mview/eval/>
- I.3 <http://grail.cs.washington.edu/projects/stfaces/>
- I.4-I.12 Adapted from Gregory, *Eye and Brain*.
- I.13 <http://xkcd.com/941/>